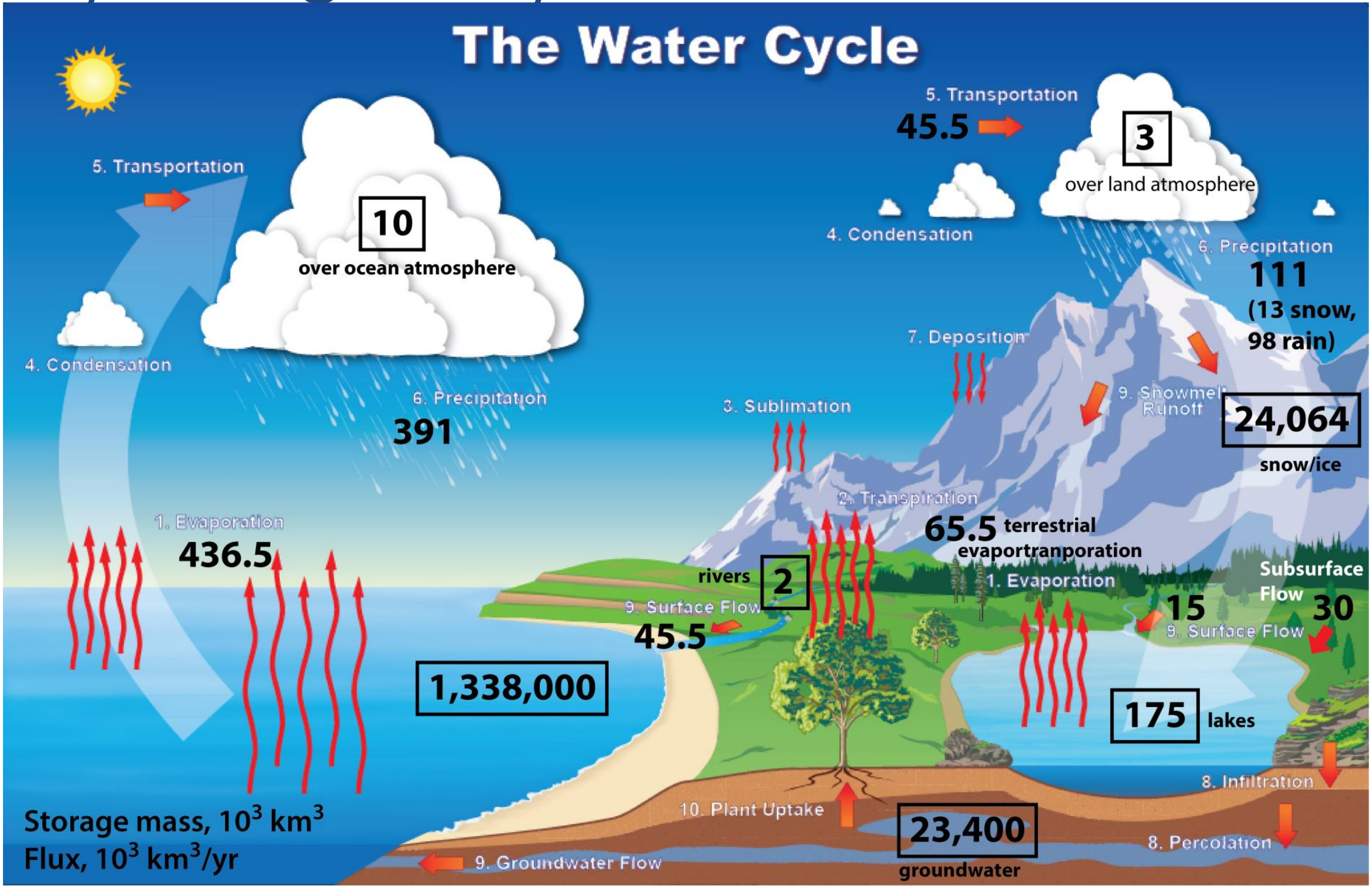
Hydrological Cycle



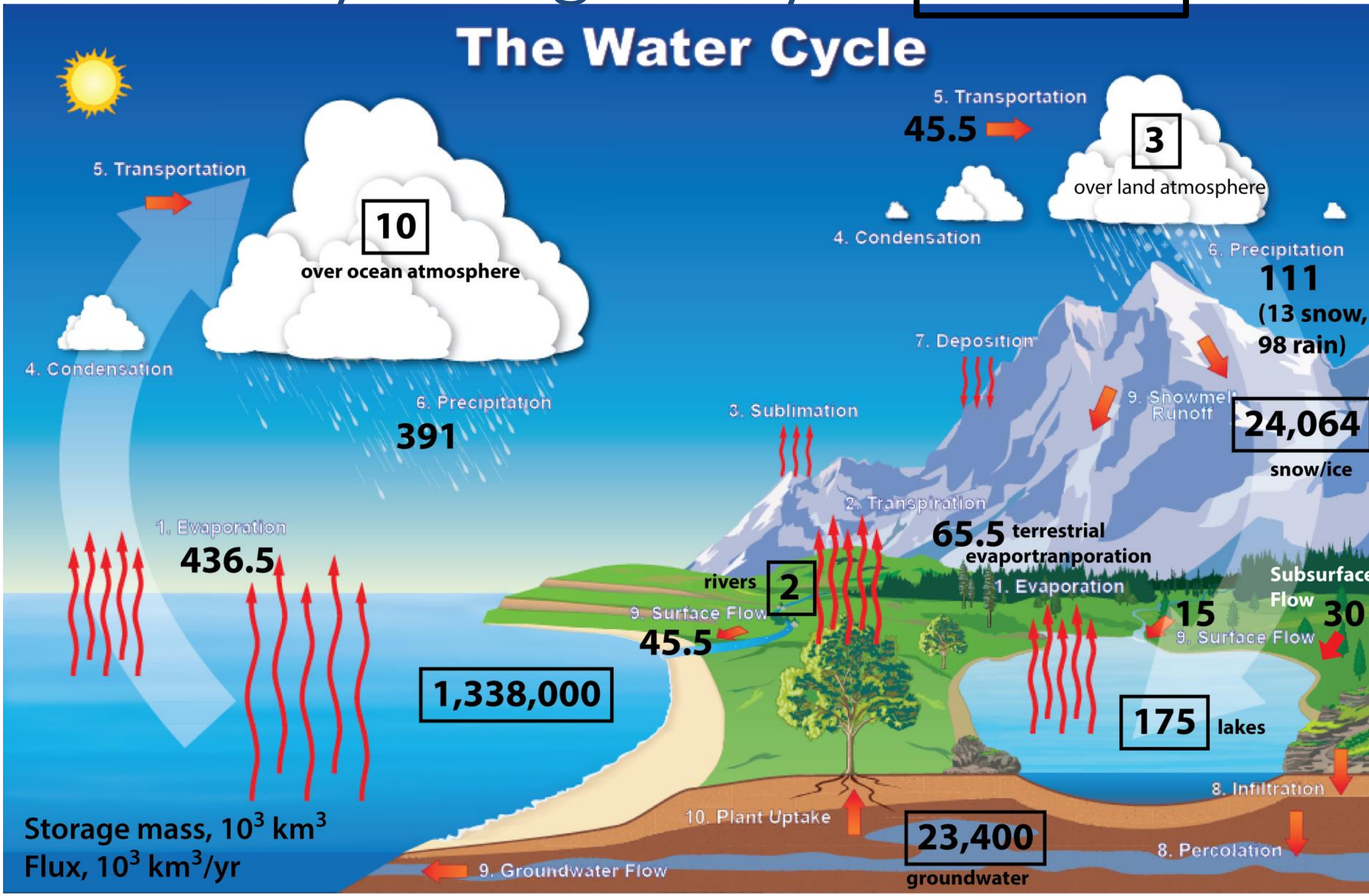
Terrestrial water balance does not include Antarctica

Water source	Water volume, in cubic miles	Water volume, in cubic kilometers	Percent of freshwater	Percent of total water
Oceans, Seas, & Bays	321,000,000	1,338,000,000	--	96.54
Ice caps, Glaciers, & Permanent Snow	5,773,000	24,064,000	68.7	1.74
Groundwater	5,614,000	23,400,000	--	1.69
Fresh	*2,526,000*	*10,530,000*	*30.1*	*0.76*
Saline	*3,088,000*	*12,870,000*	*--*	*0.93*
Soil Moisture	3,959	16,500	0.05	0.001
Ground Ice & Permafrost	71,970	300,000	0.86	0.022
Lakes	42,320	176,400	--	0.013
Fresh	*21,830*	*91,000*	*0.26*	*0.007*
Saline	*20,490*	*85,400*	*--*	*0.006*
Atmosphere	3,095	12,900	0.04	0.001
Swamp Water	2,752	11,470	0.03	0.0008
Rivers	509	2,120	0.006	0.0002
Biological Water	269	1,120	0.003	0.0001

Source: Igor Shiklomanov's chapter "World fresh water resources" in Peter H. Gleick (editor), 1993, Water in Crisis: A Guide to the World's Fresh Water Resources (Oxford University Press, New York).

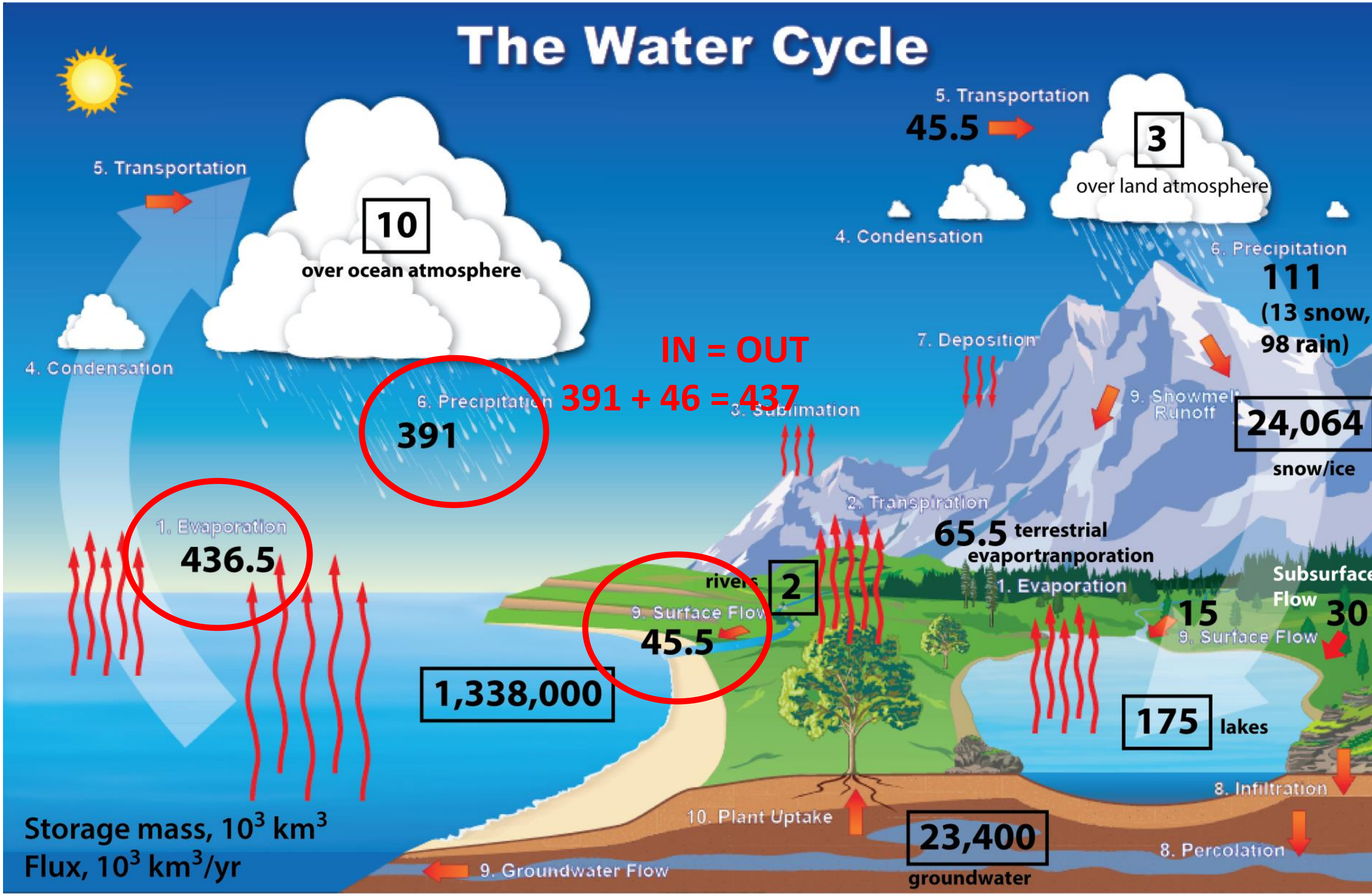
Annual Hydrological Cycle

Reservoirs



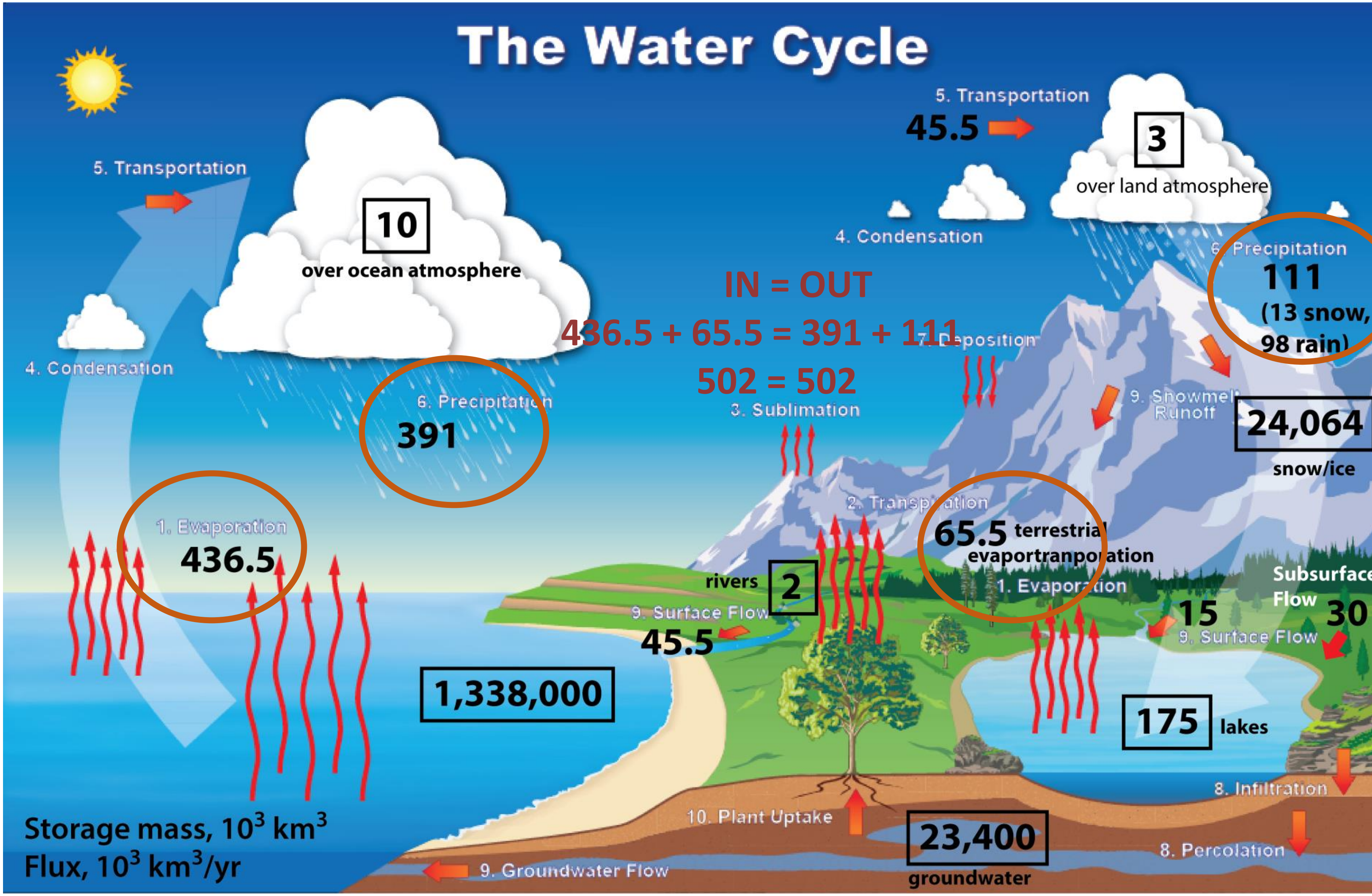
Terrestrial water balance does not include Antarctica

Annual Fluxes for Ocean Reservoir



Terrestrial water balance does not include Antarctica

Annual Fluxes for Atmospheric Reservoirs



Terrestrial water balance does not include Antarctica

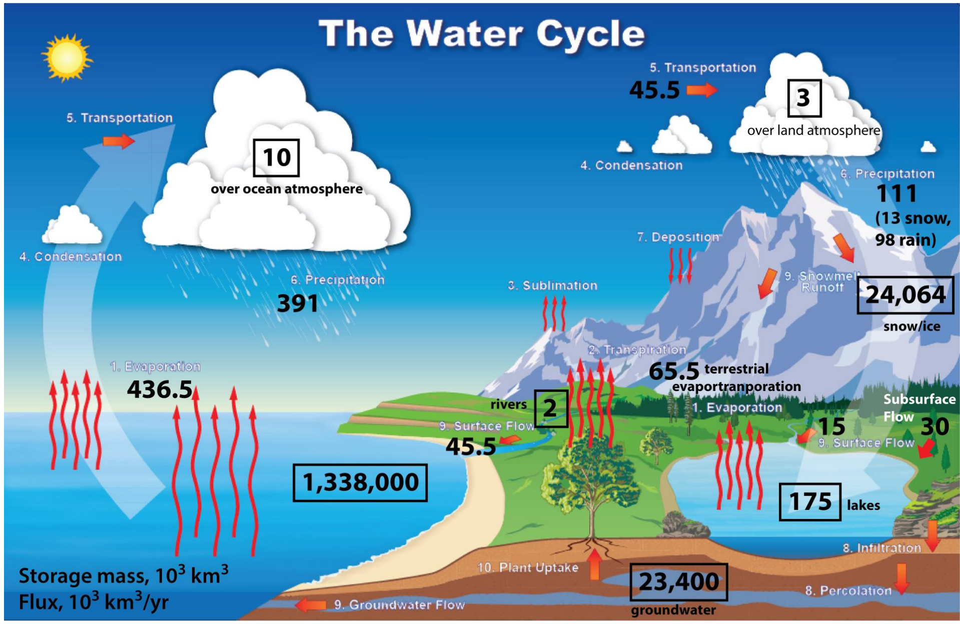
Residence Time for water in the Oceans

$$t_r = \frac{\text{Amount of water in the ocean (km}^3\text{)}}{\text{Flux of water through ocean (km}^3\text{/yr)}}$$

$$t_r = \frac{1{,}338{,}000 \times 10^3 \text{ km}^3}{436.5 \times 10^3 \text{ km}^3\text{/yr}}$$

$$t_r = \frac{1338000}{4.365}$$

$$t_r = 3065 \text{ years}$$



Terrestrial water balance does not include Antarctica

- Unlike oil, water circulates, forming closed hydrologic cycles. The amount of water will not diminish on shorter than geological time scales.

- Given this background, how could water scarcity become a widespread reality now?

A common explanation is that even though there is a lot of water on Earth, only about 2.5% is fresh water, and because most of that water is stored as glaciers or deep groundwater, only a small amount of water is easily accessible

- **This answer is only partly correct:**
- **Rather than looking only at the stocks of water resources, assessments should concentrate mainly on the flows.**
- **The amount of water stored in all the rivers in the world is only 2000 km3, much less than the annual water withdrawal of 3800 km3 /year.**
- **A more adequate measure of water availability is the 45,500 km3 /year of annual discharge, which flows mainly through the rivers from continents to the sea.**

How Much Renewable Fresh Water Is Available?

- About 3800 km3 /year of RFWR (blue water) is currently withdrawn by human beings, and that accounts for less than 10% of the maximum available RFWR in the world

- Why should we be concerned about water scarcity when presently only 10% of maximum available blue water is used?

Reason

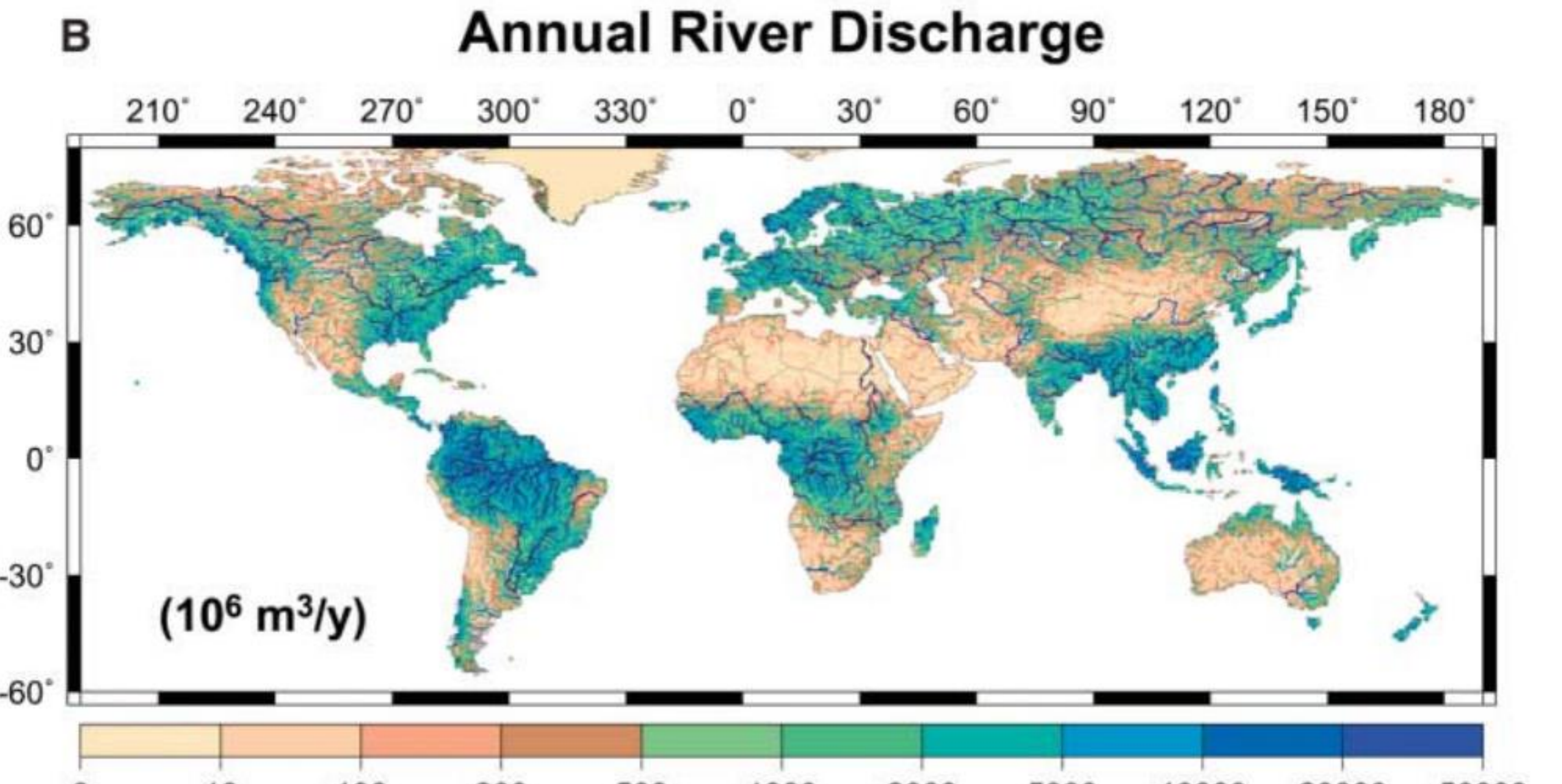
- The reason is the high variability of water resource availability in time and space.

- Time example: the monthly mean discharge in the Amazon River differs by a factor of 2 between the highest and the lowest months.

- Space example: The Ganges river has a discharge of 18,490 m^3/s. The Krishna river has a discharge of 1,642 m^3/s.

- nearly zero in desert areas through more than 2000 mm/year of runoff in the tropics and more than 200,000 m3 /s of discharge on average near the river mouth of the Amazon.

- Because of this temporal and spatial variability, it is impractical to use 100% of the available RFWR for human society. Flow during floods and wet seasons cannot be used during the low flow seasons unless storage systems are in place.



The water scarcity index

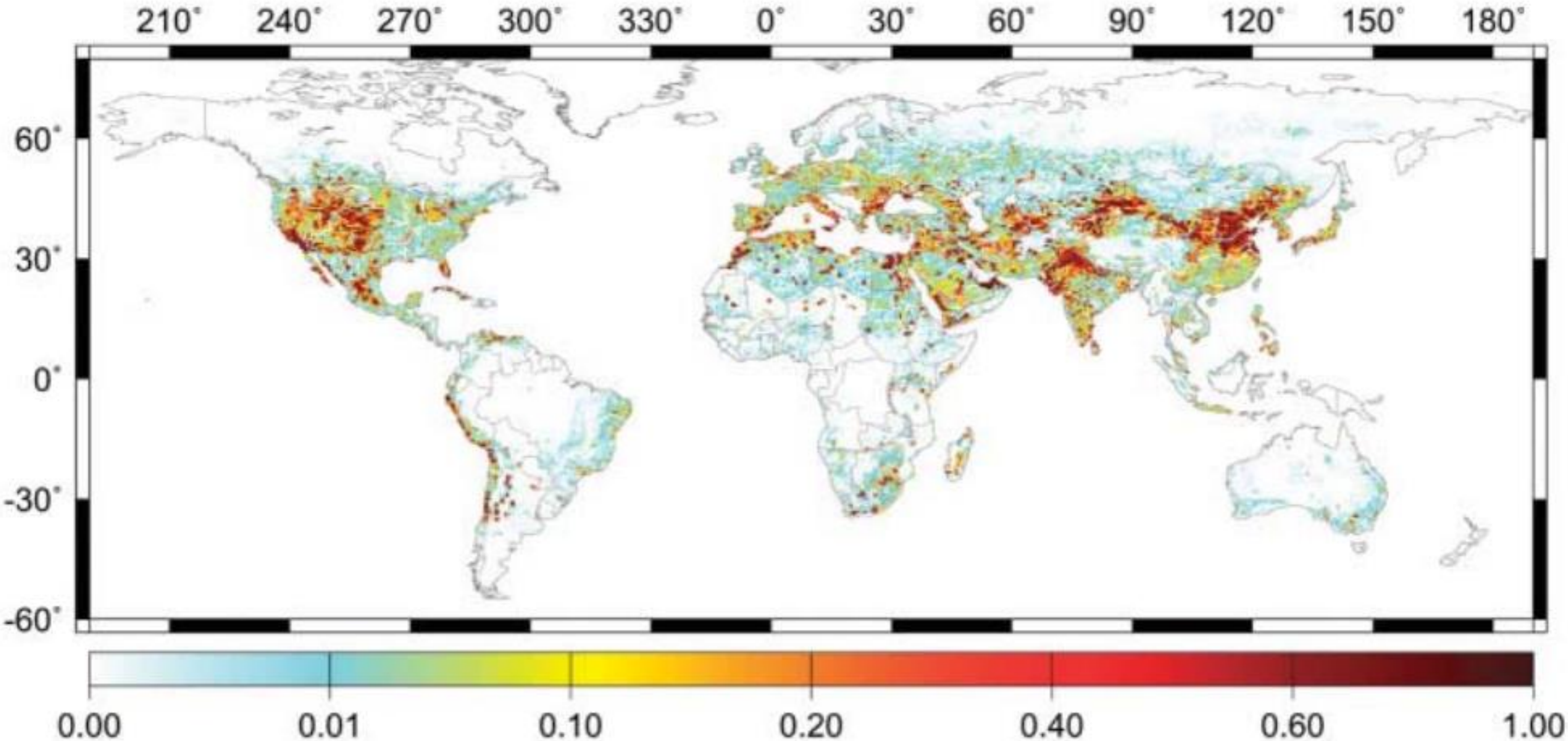
- The water scarcity index is defined as

$$Rws = (\mathbf{W} - \mathbf{S})/\mathbf{Q},$$

- where W = **annual water withdrawal by all the sectors**, S = **the water use from desalinated water**, and Q = **annual RFWR**, respectively.

- A region is usually considered highly water stressed if Rws is higher than 0.4

c **Water Scarcity Index**



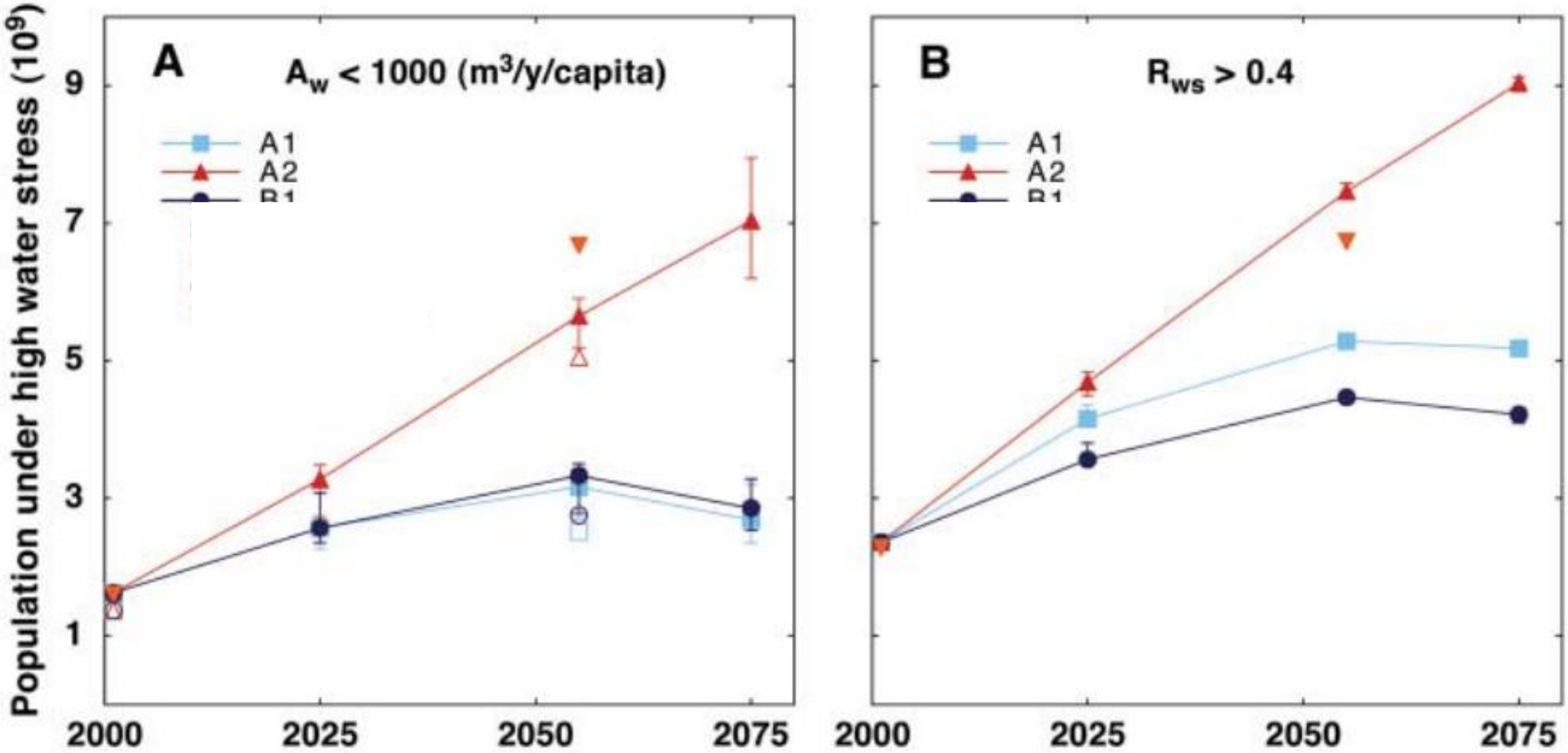
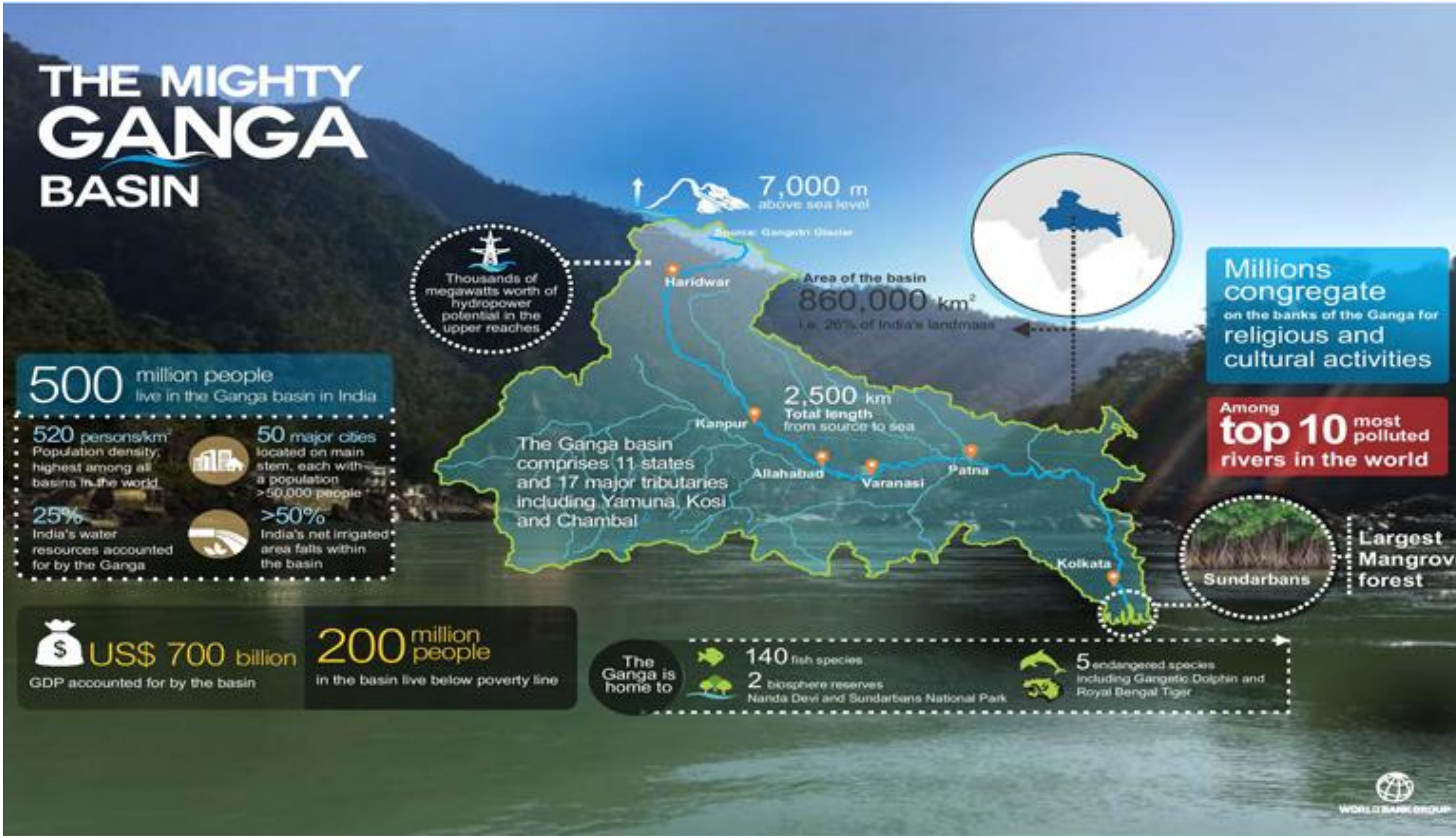
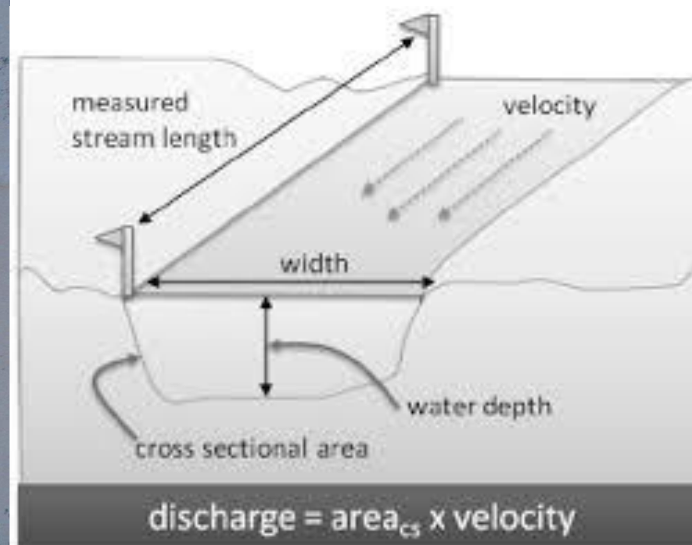


Fig. 3. Current and future projections of population under high water stress under three business-as-usual scenarios of the Intergovernmental Panel on Climate Change's Special Report on Emissions Scenarios. Threshold values are set to be (**A**) the water-crowding indicator $Aw = Q/C < 1000$ m^3/year per capita and (**B**) the water scarcity index $Rws = (W - S)/Q > 0.4$, where Q, C, W, and S are renewable freshwater resources (RFWR), population, water withdrawal, and water generated by desalination, respectively. Error bars indicate the maximum and minimum population under high water stress corresponding to the RFWR projected by six climate models. Climatic conditions averaged for 30 years are used for the plots at 2025 (averaged for 2010 to 2039), 2055 (averaged for 2040 to 2069), and 2075 (averaged for 2060 to 2089).

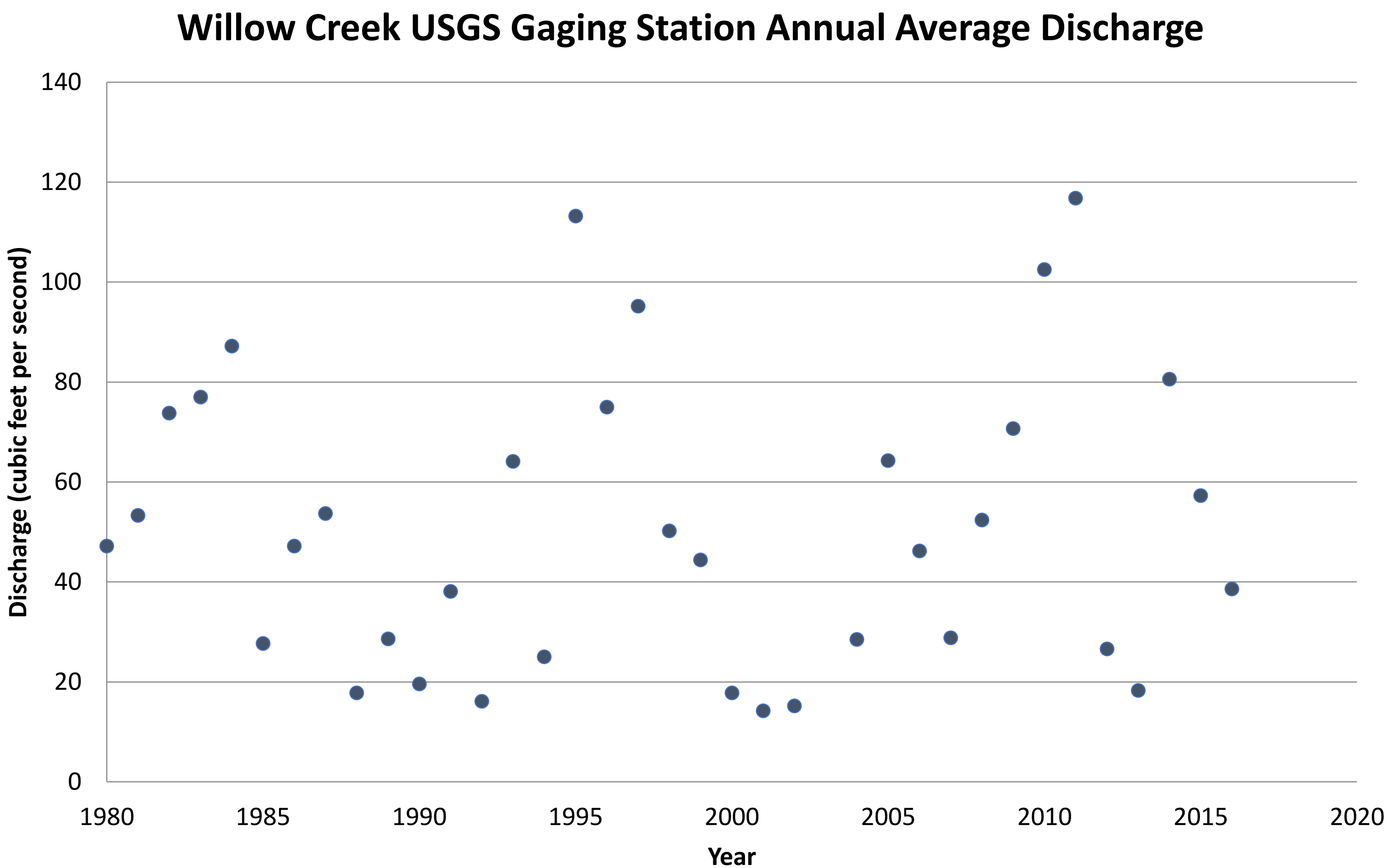
Traditional Hydrologic Measurements


CHINA
NEPAL
BHUTAN
BANGLADESH
INDIA
MYANMAR
Lhasa
New Delhi
Jaipur
Agra
Lucknow
Kanpur
Kathmandu
Allahabad
Varanasi
Patna
Farakka
Guwahati
Dhaka
Kolkata
Indore
Bhopal
Yarlung Zangbo (Tsangpo)
Subansiri
Lohit
Brahmaputra
Teesta
Jamuna
Surma
Barak
Meghna
Padma
Hooghly
Damodar
Ganges
Yamuna
Chambal
Banas
Betwa
Gomti
Ghaghara
Gandak
Kosi
Tamsa
Son
Mouths of the Ganges
Bay of Bengal

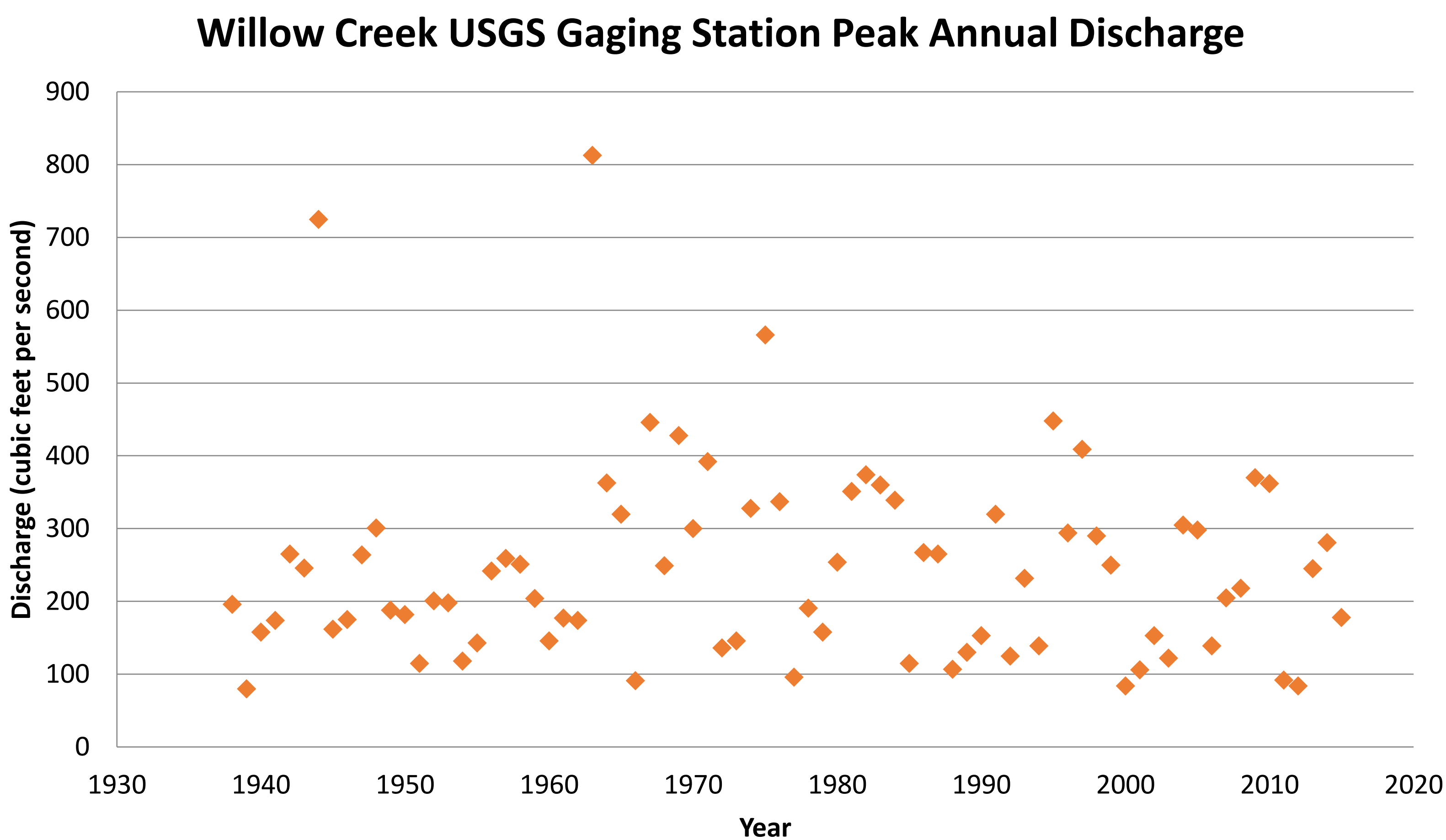

THE MIGHTY
GANGA
BASIN
7,000 m
above sea level
Source: Gangotri Glacier
Thousands of megawatts worth of hydropower potential in the upper reaches
Haridwar
Area of the basin
860,000 km²
i.e. 26% of India's landmass
Millions congregate
on the banks of the Ganga for
religious and cultural activities
Among
top 10 most polluted
rivers in the world
500 million people
live in the Ganga basin in India
520 persons/km²
Population density, highest among all basins in the world
50 major cities
located on main stem, each with a population >50,000 people
25%
India's water resources accounted for by the Ganga
>50%
India's net irrigated area falls within the basin
2,500 km
Total length
from source to sea
Kanpur
The Ganga basin comprises 11 states and 17 major tributaries including Yamuna, Kosi and Chambal
Allahabad
Varanasi
Patna
Kolkata
Sundarbans
Largest Mangrove forest
US$ 700 billion
GDP accounted for by the basin
200 million people
in the basin live below poverty line
The Ganga is home to
140 fish species
2 biosphere reserves
Nanda Devi and Sundarbans National Park
5 endangered species
including Gangetic Dolphin and Royal Bengal Tiger
WORLD BANK GROUP


measured stream length
velocity
width
water depth
cross sectional area
discharge = area$_{cs}$ x velocity

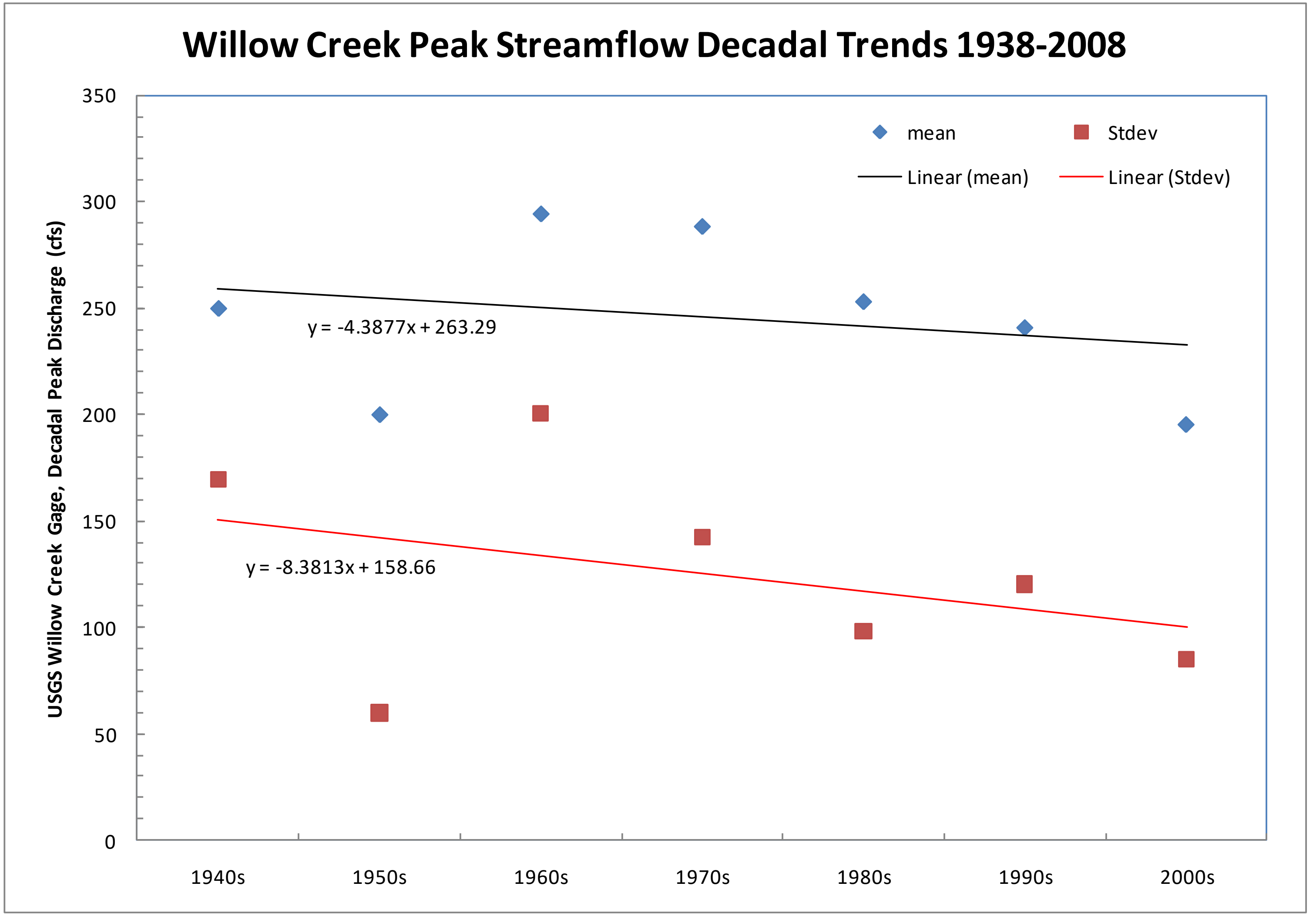
USGS Willow Creek: Average Annual



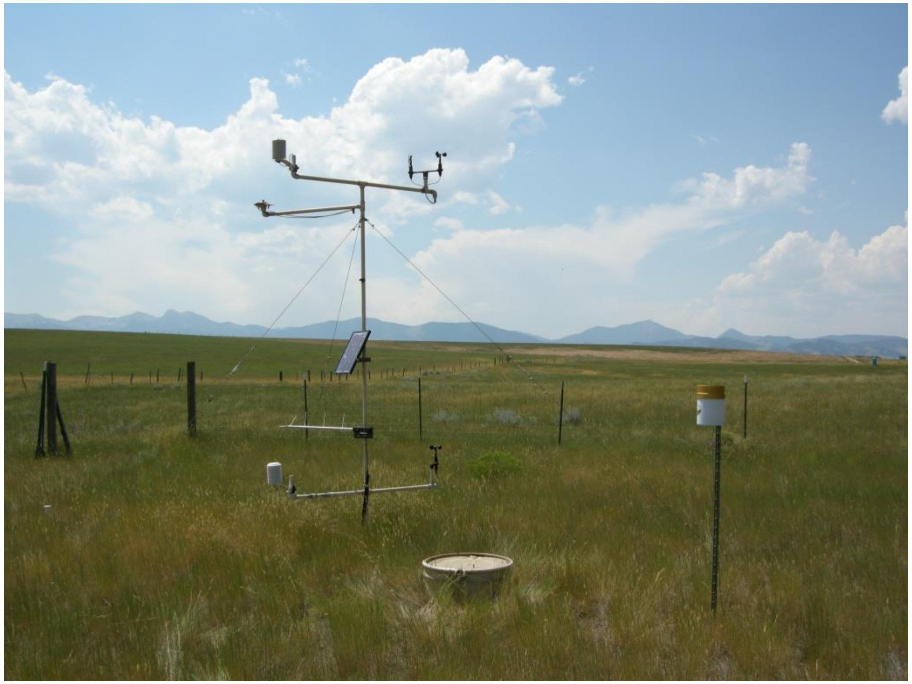
USGS Willow Creek: Annual Peak Discharge



Decrease in Decadal Peak Streamflow



Meteorological Station



Groundwater Wells (Alluvial)

Sensor in your water well

A sensor is placed above the the pump in a borehole or at the bottom of a dug well.
The sensor measures the pressure exerted by the water column above it and converts it to water level.

Connected with the Aqvify box

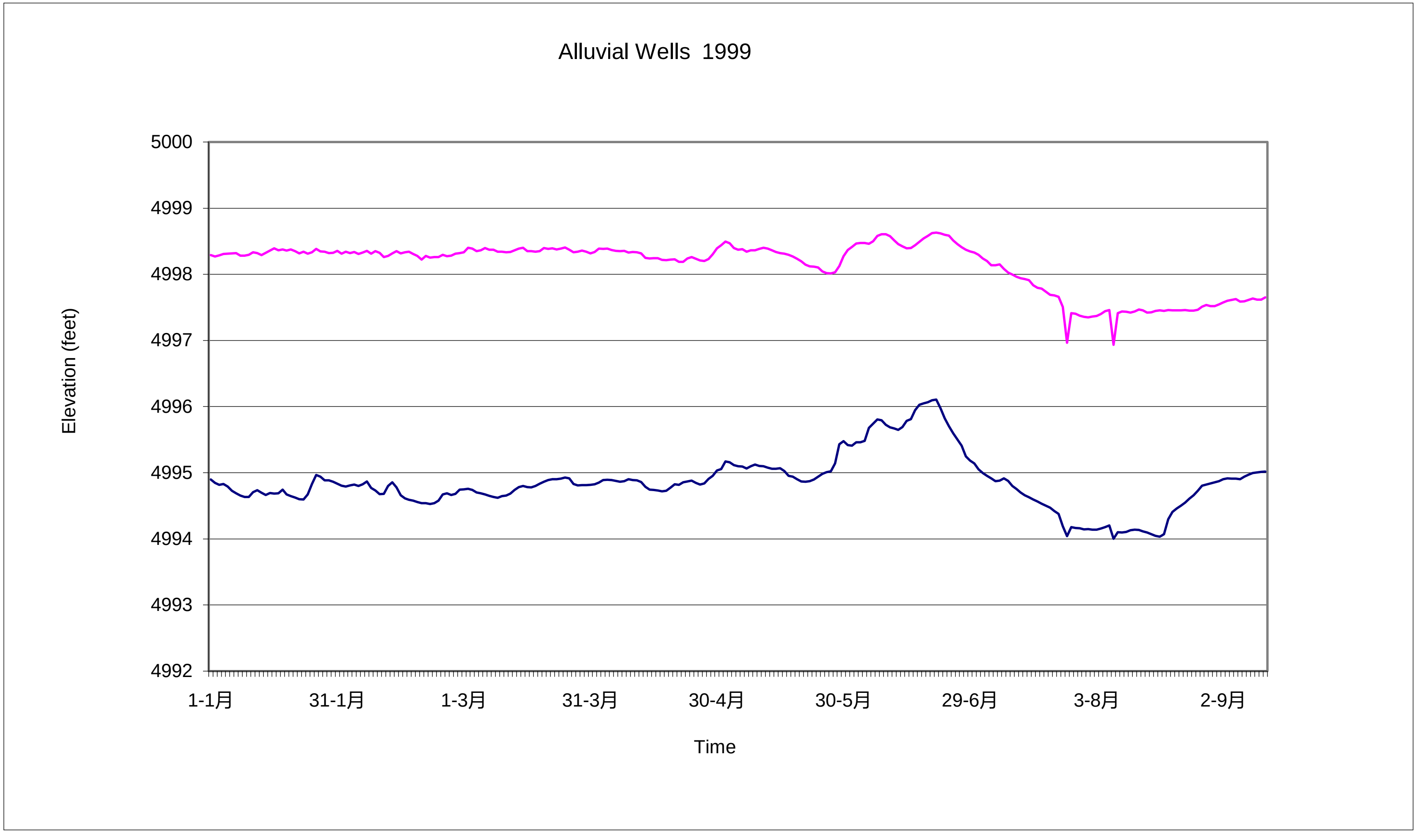
The sensor is connected to the Aqvify box. Data is sent every minute over the mobile network – SIM & data included.

Your water well in the app

See real-time status and past history in our app. Together with smart alarms and more! More about the app here.

▶ View Video Demo

Example of Groundwater record (2 STATIONS)



Groundwater

Groundwater ---> water stored in open spaces formed from atmospheric precipitation (meteoric water)within underground rocks and sediments.

The enormous reservoir of groundwater stored beneath Earth’s surface equals about 29 percent of all the fresh water stored in lakes and rivers, glaciers and polar ice, and the atmosphere.

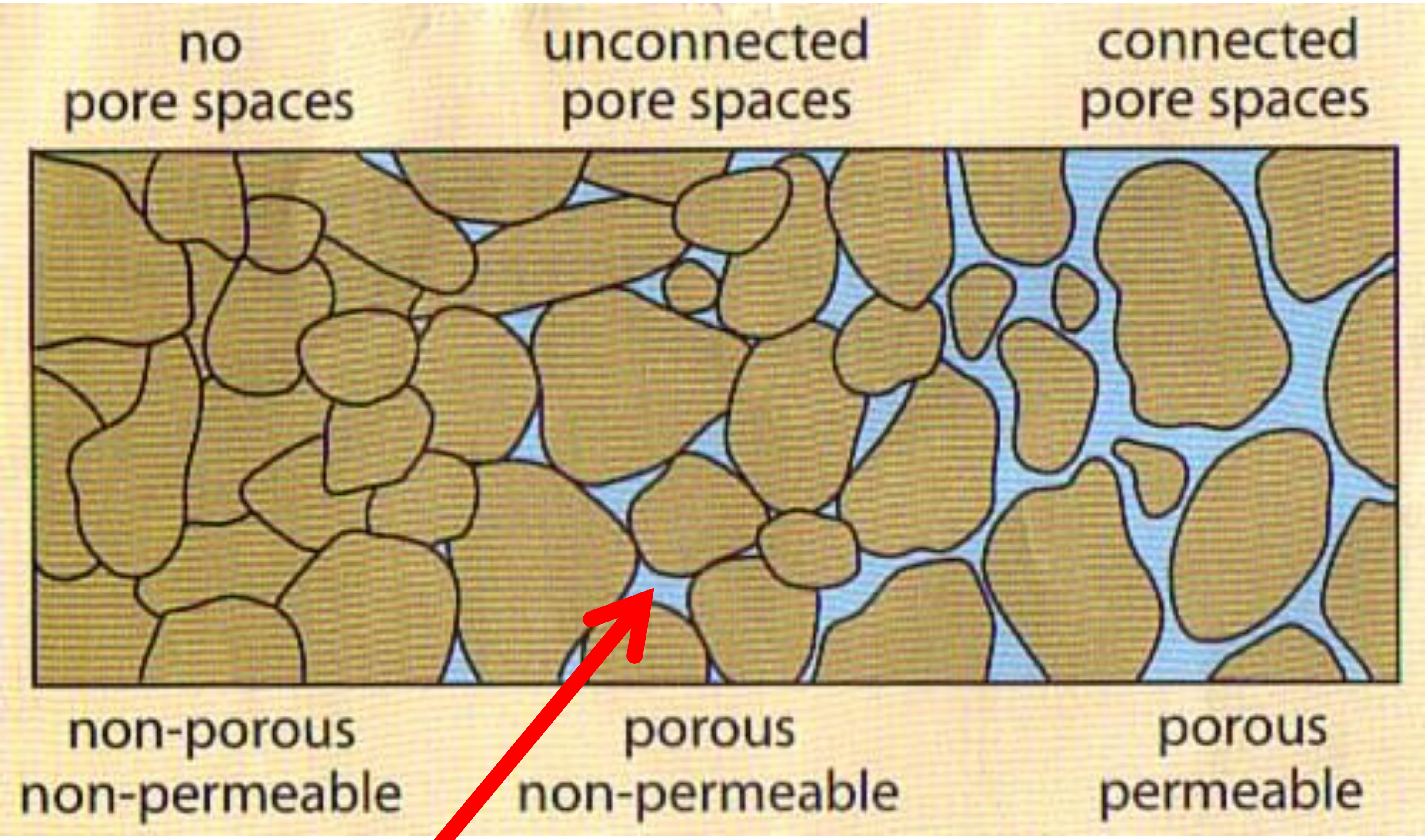
Vocabulary

Porosity & Permeability

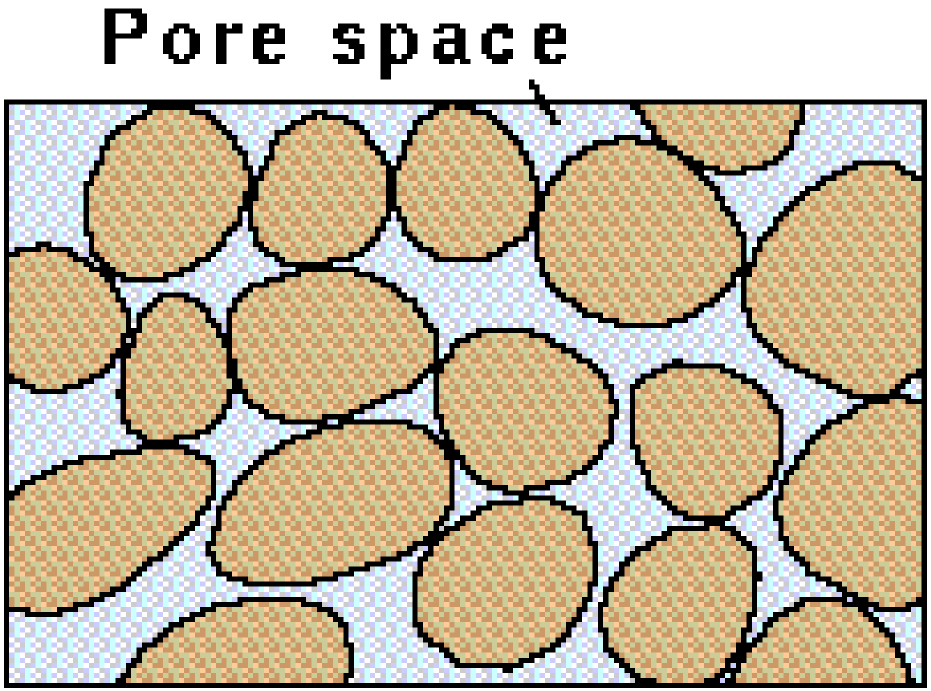
- Porosity
  > that portion of the material's volume which is **pore spaces.**

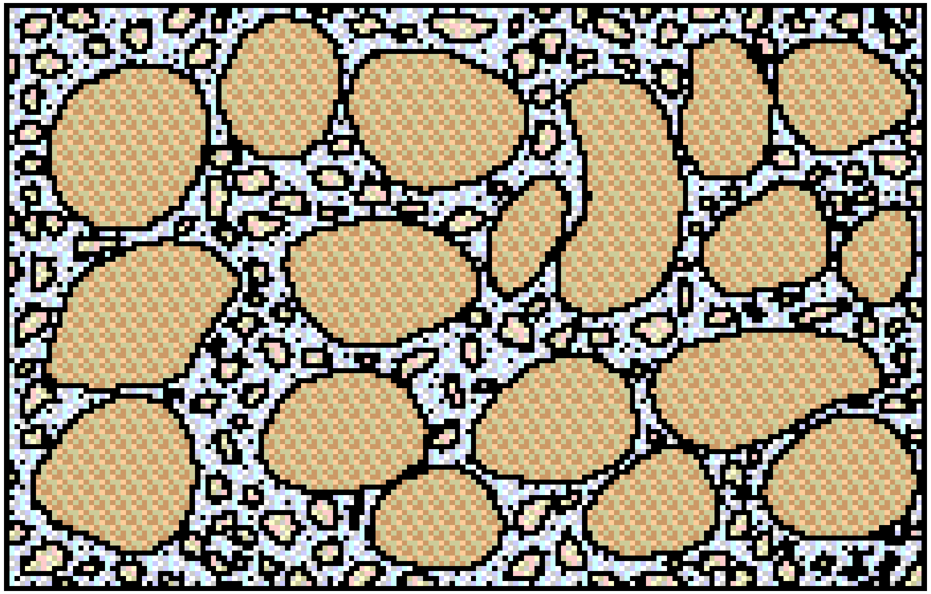
  > i.e. ***the amount of empty space in rock or sediments***
- Permeability
  > the ability of a material to transmit fluids.

  > i.e. *the degree of interconnectedness of the empty space*



Notice that the pore spaces do not connect!





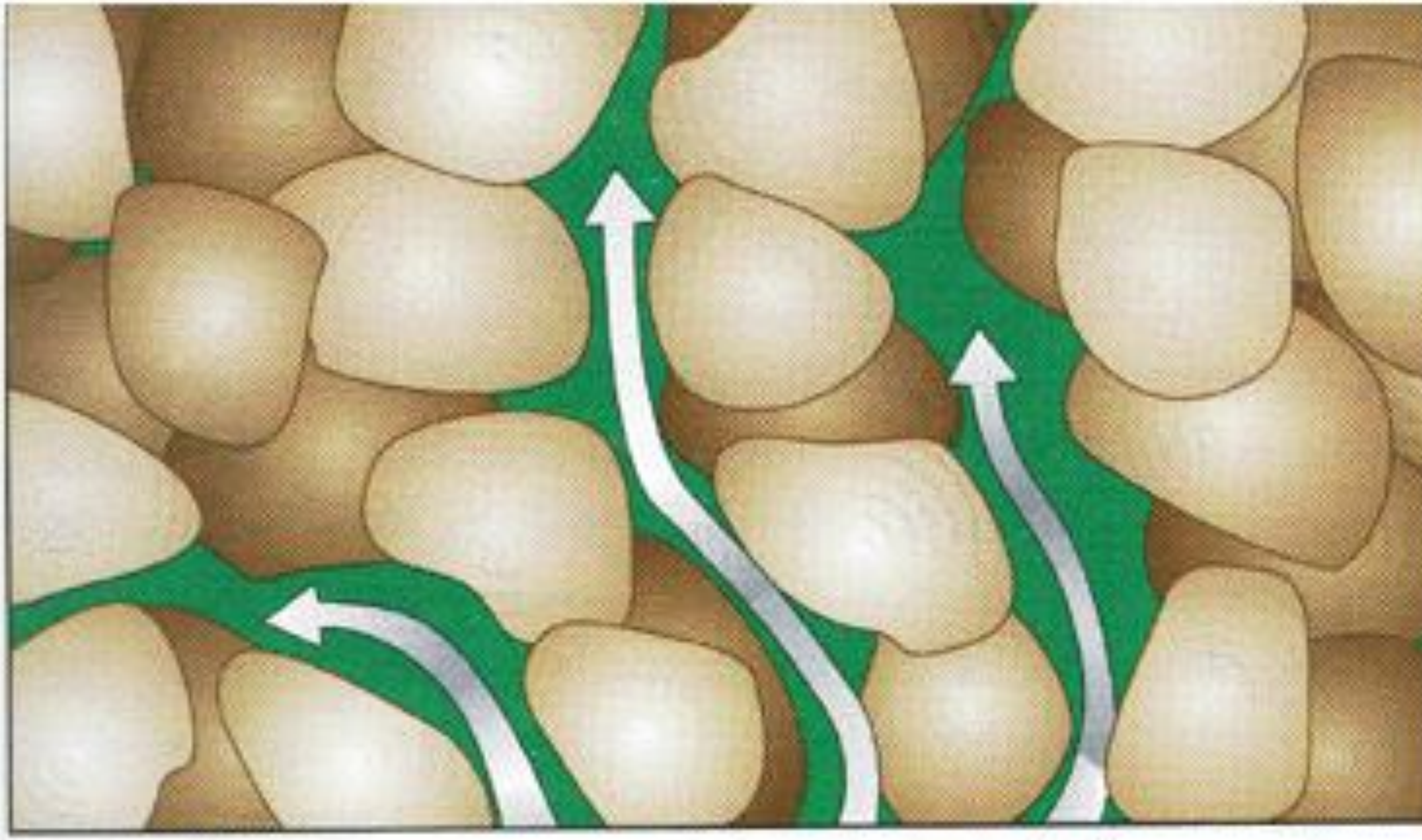
Porosity affected by:

- particle size,
- particle shape,
- sorting of particles,
- amount of material cementing the grains together,
- fractures in materials,
- and internal arrangement of particles

There are three types of pores:
spaces between grains (intergranular porosity),
spaces in fractures (fracture porosity),
and spaces created by dissolution (vuggy porosity).

Porosity is lower in igneous and metamorphic rocks, in which pore space is created mostly by fractures, including
joints and cleavage at natural zones of weakness

Pore space in limestones and other highly soluble rocks such as evaporites may be created when groundwater interacts with the rock and partly dissolves it, leaving
irregular voids known as vugs.



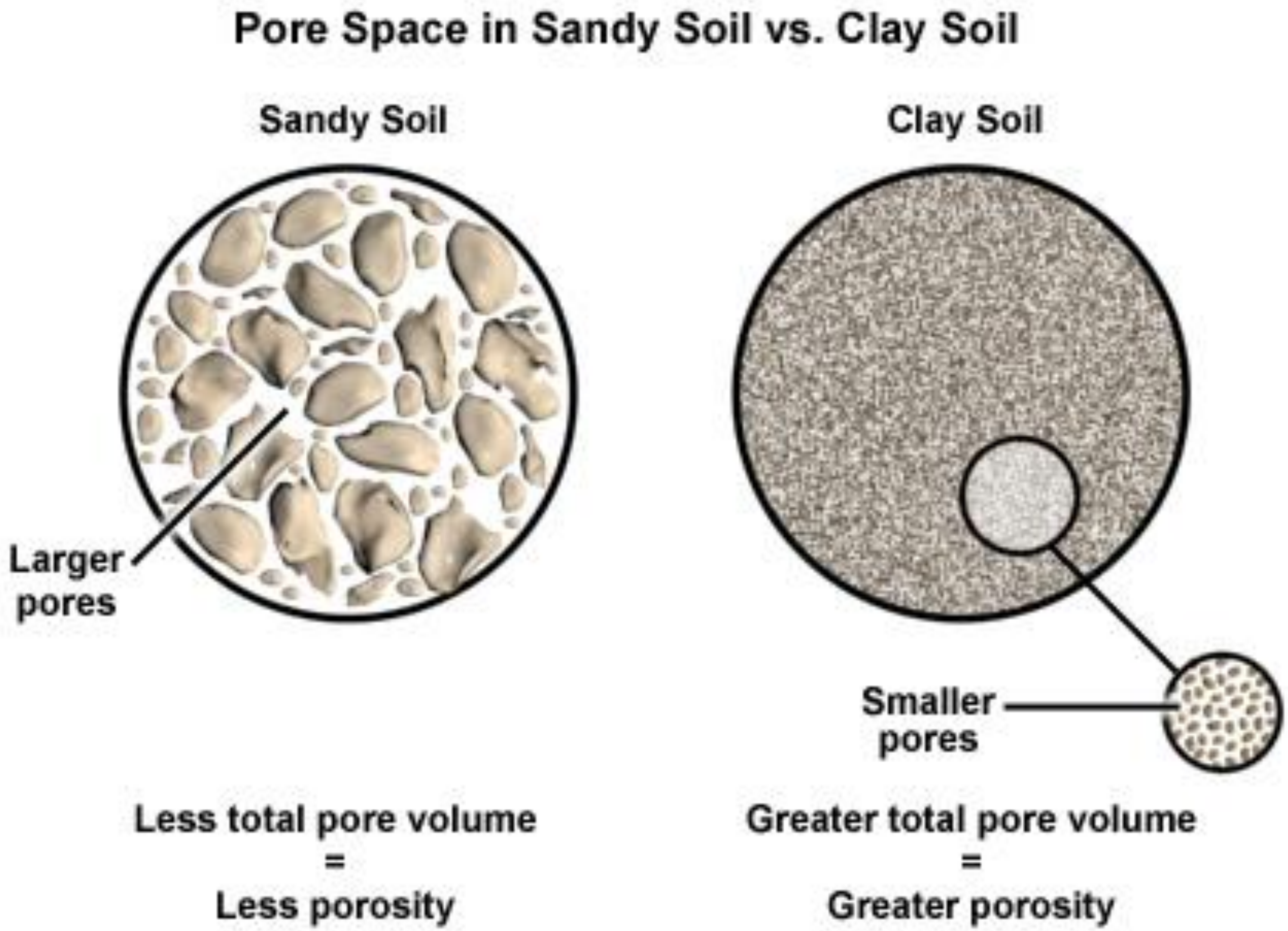
Connected pores give a rock permeability.

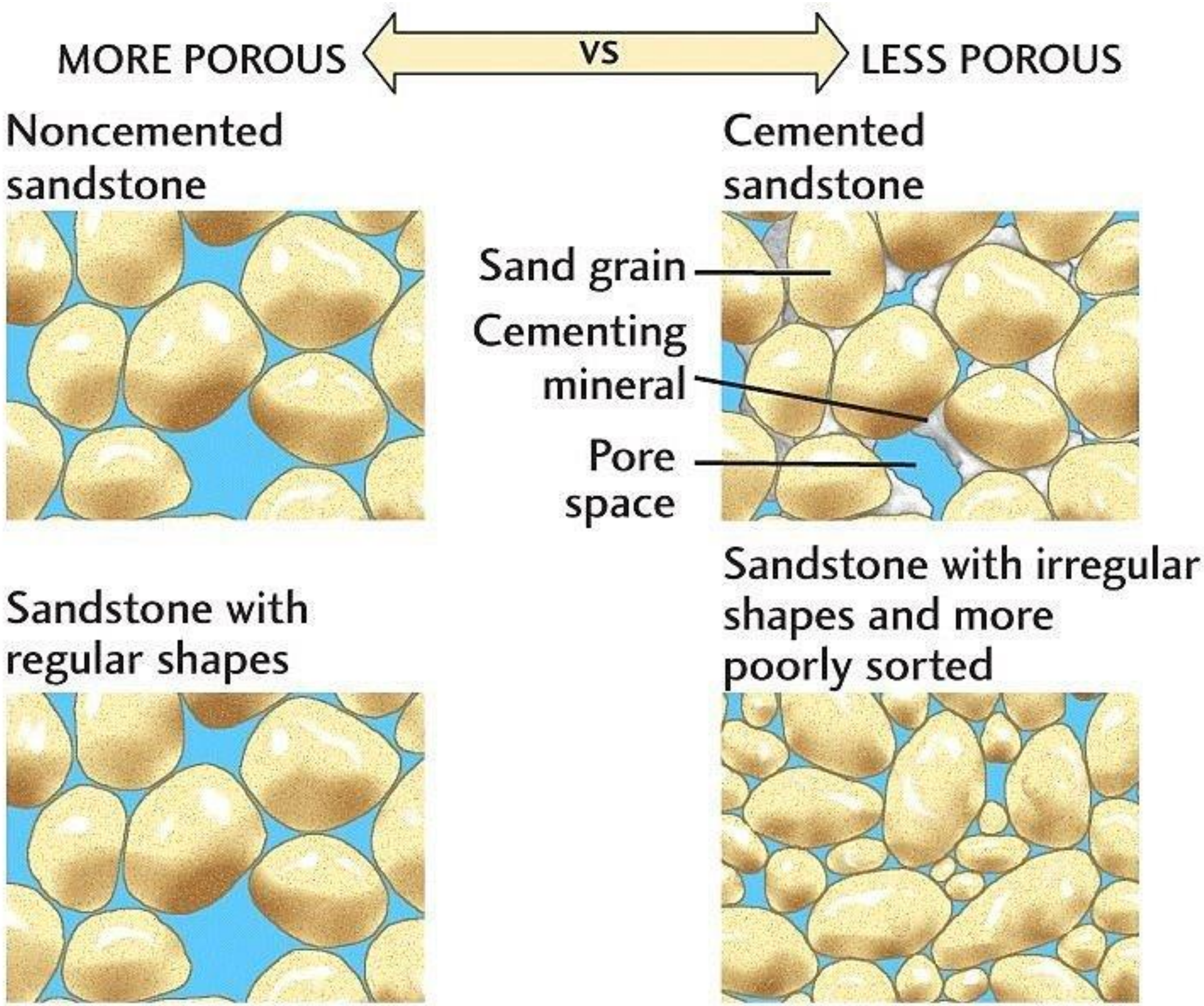
Connections between pore spaces are wider in coarse-grained material (gravels) than fine-grained material (sand).

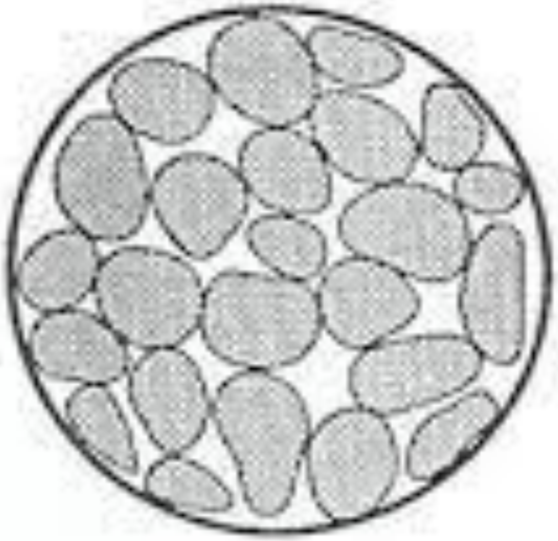
Typical Porosity values

Material	% Porosity
Unconsolidated sediment	
Soil	55
Gravel	20-40
Sand	25-50
Silt	35-50
Clay	50-70

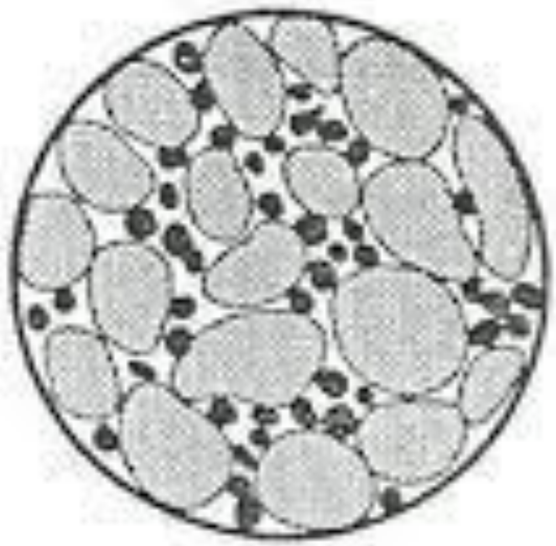
Material	% Porosity
Rocks	
Sandstone	5-30
Shale	0-10
Solution in limestone, dolostone	10-30
Fractured basalt	5-40
Fractured granite	~10


Pore Space in Sandy Soil vs. Clay Soil
Sandy Soil
Clay Soil
Larger pores
Smaller pores
Less total pore volume
=
Less porosity
Greater total pore volume
=
Greater porosity


MORE POROUS
VS
LESS POROUS
Noncemented sandstone
Cemented sandstone
Sand grain
Cementing mineral
Pore space
Sandstone with regular shapes
Sandstone with irregular shapes and more poorly sorted



Sand—
high porosity



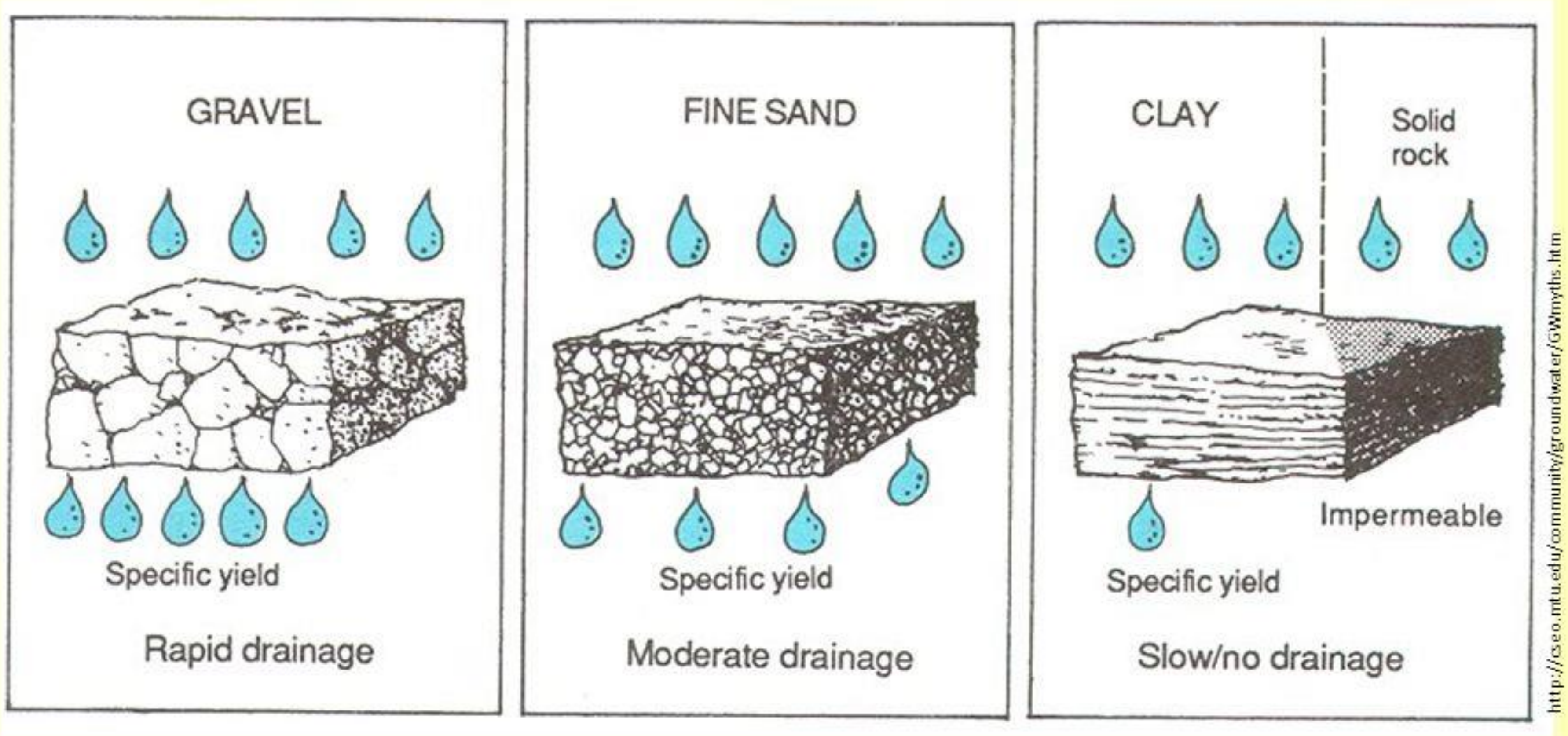
Sand and silt—
lower porosity

Permeability

The capacity of a solid to allow fluids to pass through it is its Permeability

- **Materials must have both Porosity & Permeability to allow water to move through it!**

Generally, permeability increases as porosity increases, but permeability also depends on the sizes of the pores, how well they are connected, and how is path the fluid must travel to pass through the material.

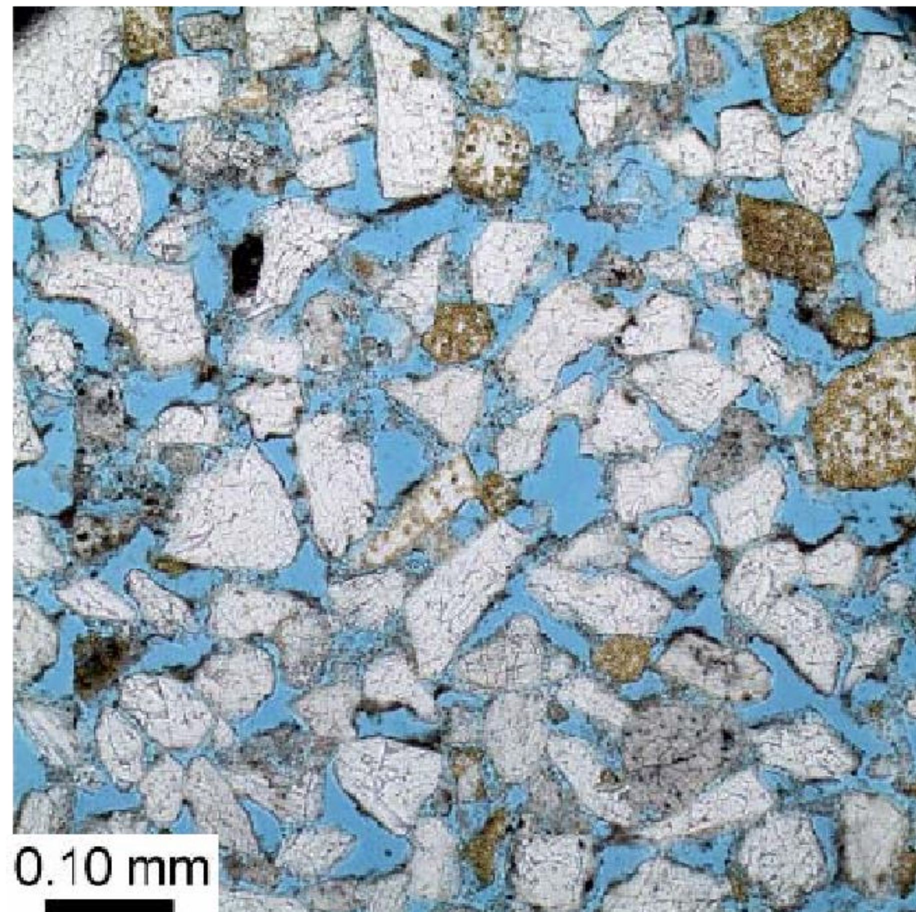


Each type of soil or rock has a level of permeability, which is a measure of the speed at which water moves through porous material. For example, gravel and sand have higher permeability than clay. Some flow paths are also longer than others.

http://techalive.mtu.edu/meec/module06/Permeability.htm

Materials with high porosity <u>*AND*</u> high permeability

- Unlithified (sediment)
  - Sand
  - Gravel
- Lithified (rock)
  - Sandstone
  - fractured limestone
  - fractured granites



Rock or Sediment Type	Porosity	Permeability
Gravel	Very high	Very high
Coarse- to medium-grained sand	High	High
Fine-grained sand and silt	Moderate	Moderate to low
Sandstone, moderately cemented	Moderate to low	Low
Fractured shale or metamorphic rock	Low	Very low
Unfractured shale	Very low	Very low

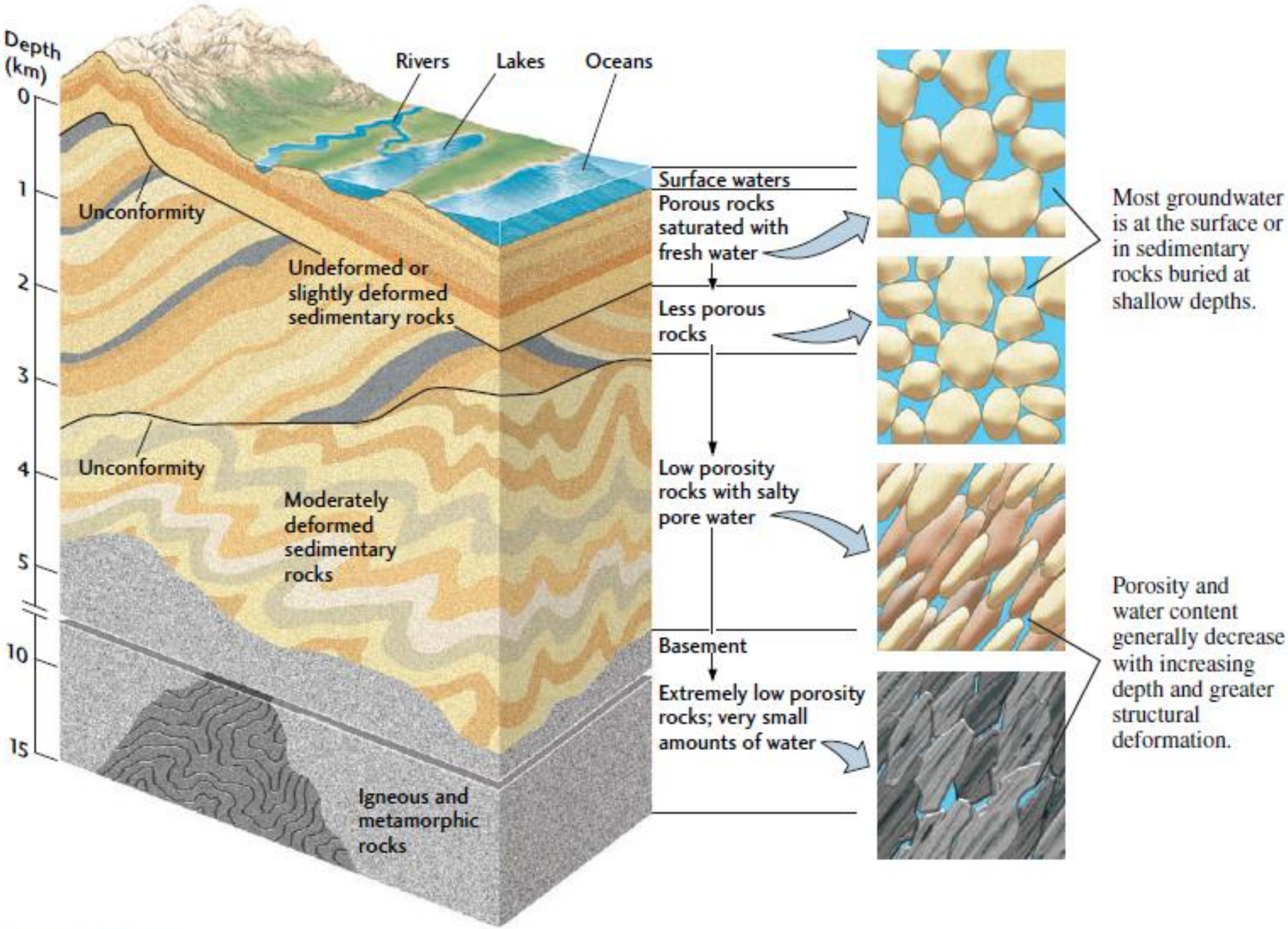
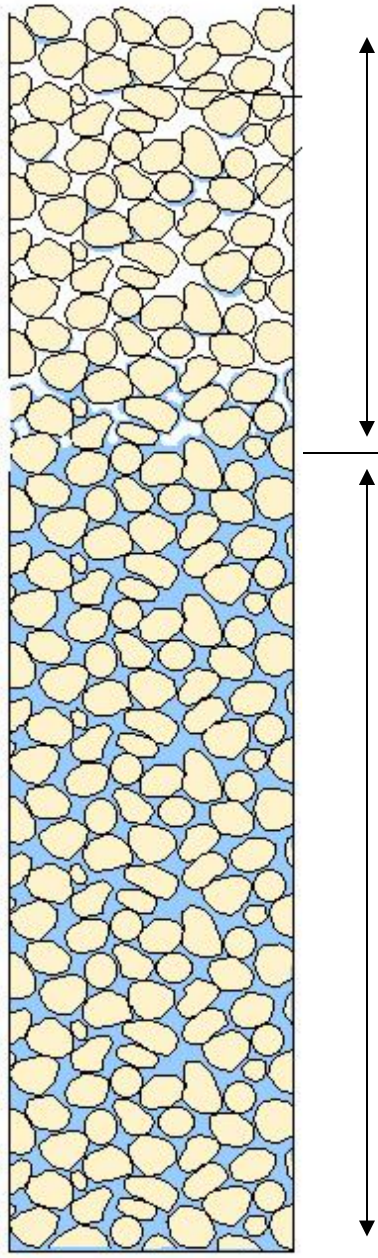


FIGURE 17.25 ■ Porosity and permeability, and therefore water content, generally decrease with increasing depth in Earth's crust.

The Groundwater Table

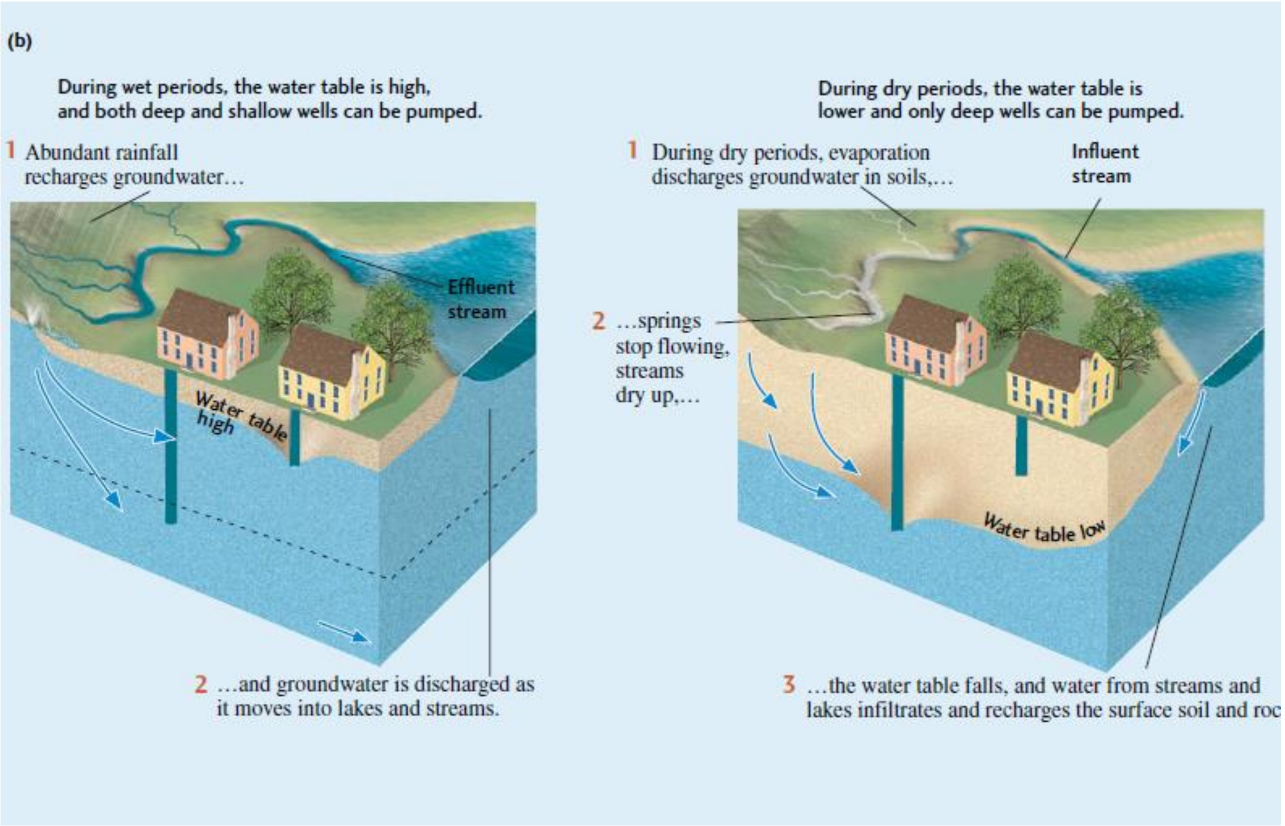


Zone of Aeration: area where the pore spaces in the rock/soil are empty of water (Unsaturated zone/vadose zone)

Water Table: boundary between zone of saturation & zone of aeration; wells must go below the water table to reach water

Zone of Saturation: area where the pore spaces in the rock/soil are filled with water (phreatic zone)

The Groundwater Table



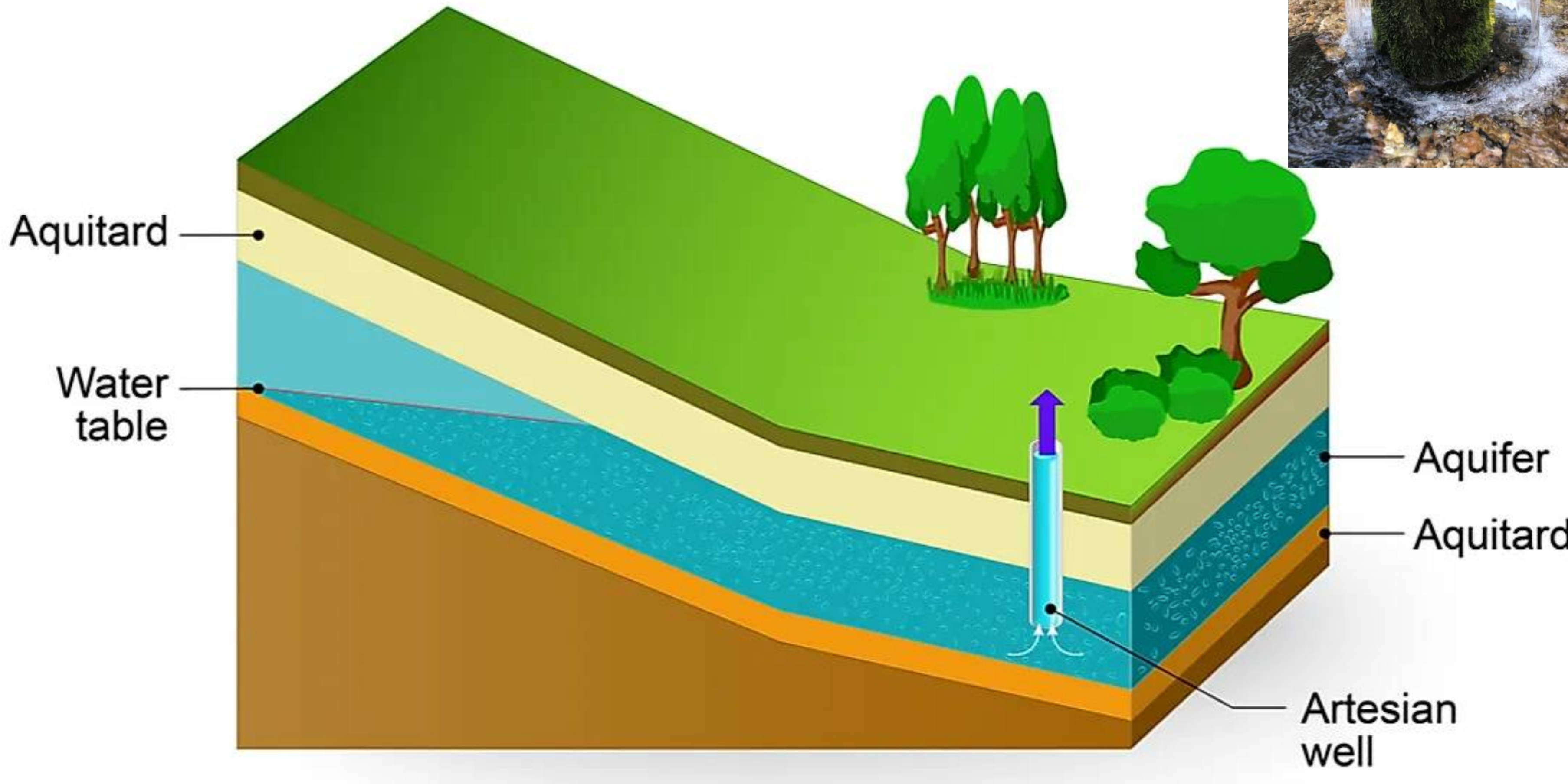
Water enters and leaves the saturated zone through recharge and discharge. Recharge is the infiltration of water into any subsurface formation. Rain and melting snow are the most common sources of recharge. Discharge,
the movement of groundwater to the surface, is the
opposite of recharge. Groundwater is discharged by evaporation,
through springs, and by pumping from artificial wells.

Aquifers and Aquicludes

Rock formations through which groundwater flows in sufficient quantity to supply wells are called aquifers.

- Aquifer
  - a porous & permeable layer capable of transmitting groundwater.
    - well rounded, well sorted, sand & gravel.

- Aquiclude (or Aquitard)
  - an impermeable layer preventing movement of groundwater.
    - Shales, unfractured Ign & Mm rock, crystalline limestone.

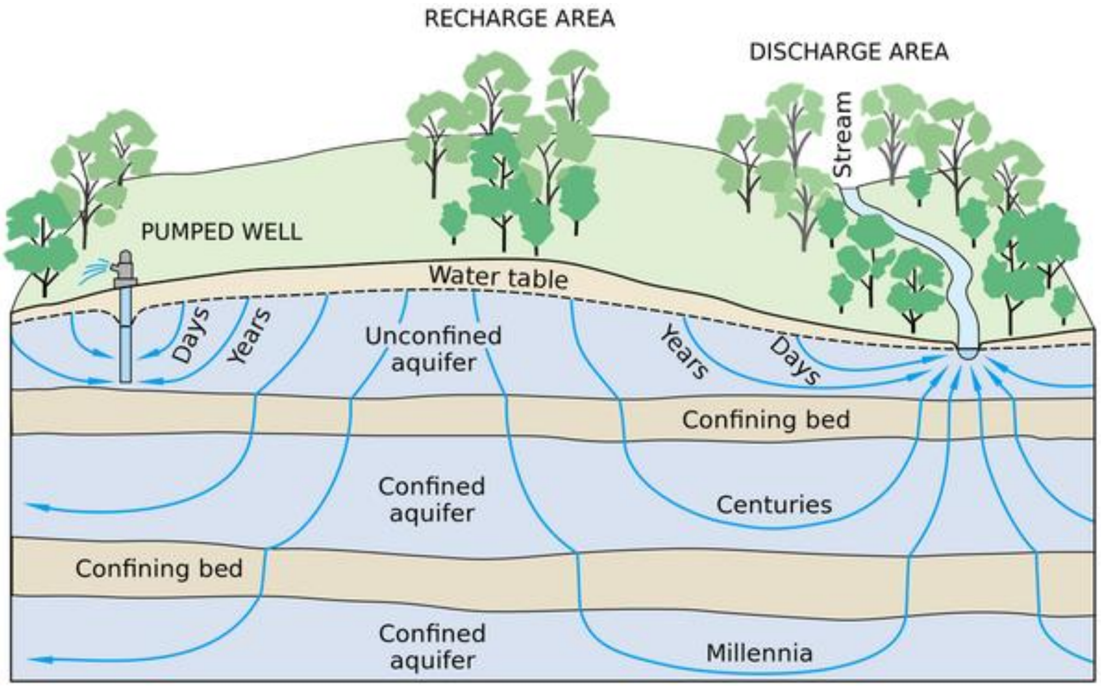


well from which water flows under natural pressure without pumping

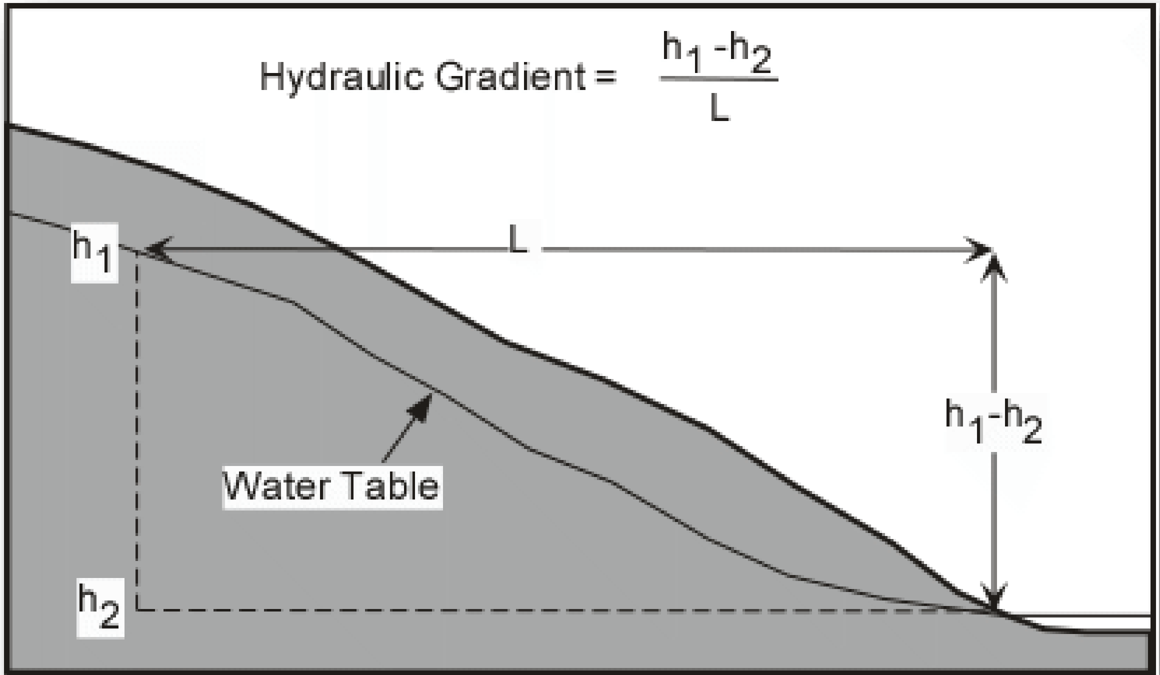
Unconfined aquifers & Confined aquifers

Unconfined aquifers are where the rock is directly open at the surface of the ground and groundwater is directly recharged, for example by rainfall or snow melt.

Confined aquifers are where thick deposits overly the aquifer and confine it from the Earth's surface or other rocks. Confining beds or cover, such as clay or unfractured mudstones prevent or slow groundwater movement



Groundwater flows due to differences in elevation and water pressure.



Stephen A. Nelson, Tulane University

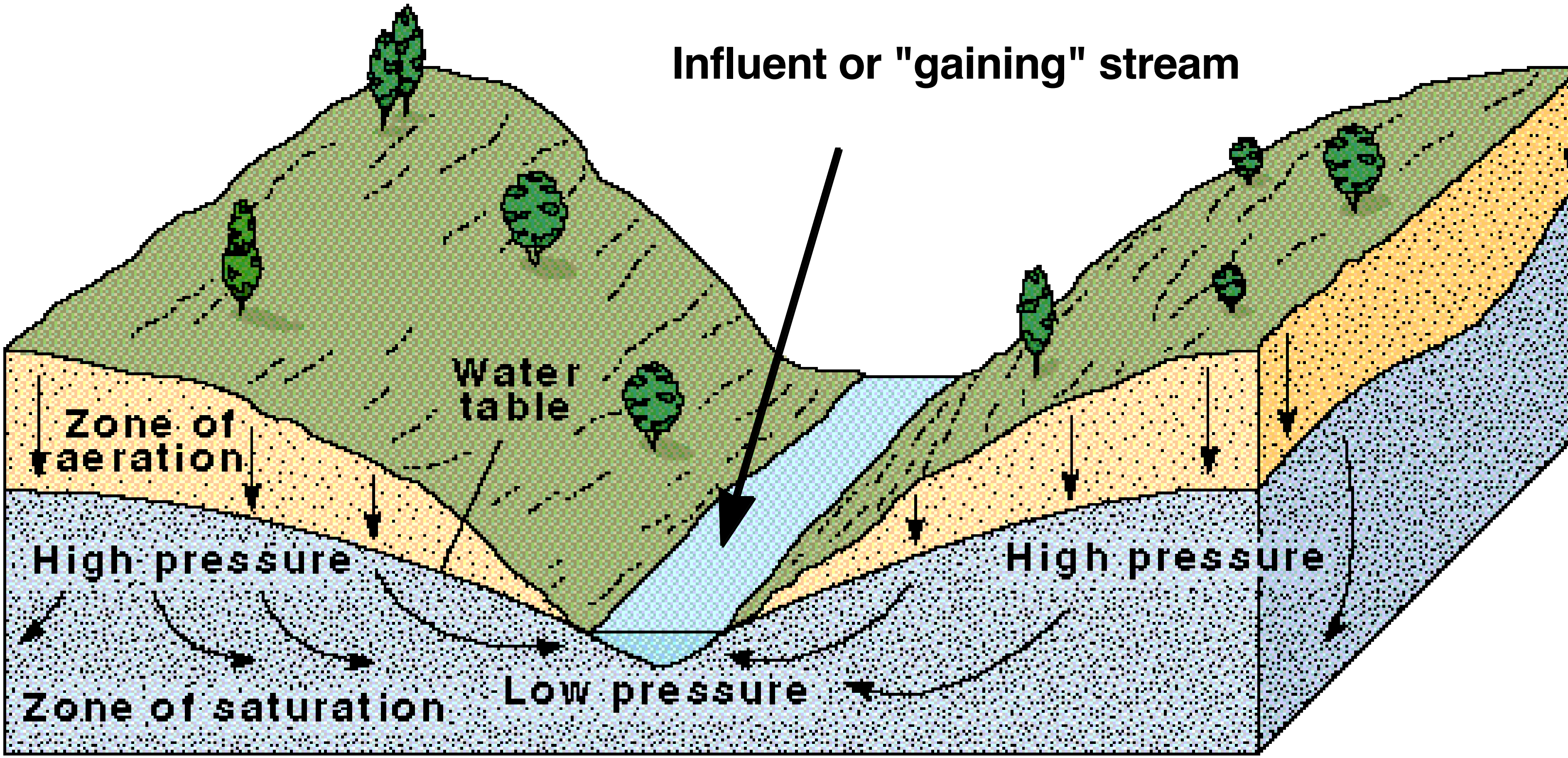
Hydraulic gradient is the slope or steepness of the *water table* (unitless number).

hydraulic gradient = Δh/distance

Groundwater always flows from a region of higher *hydraulic head* to lower.

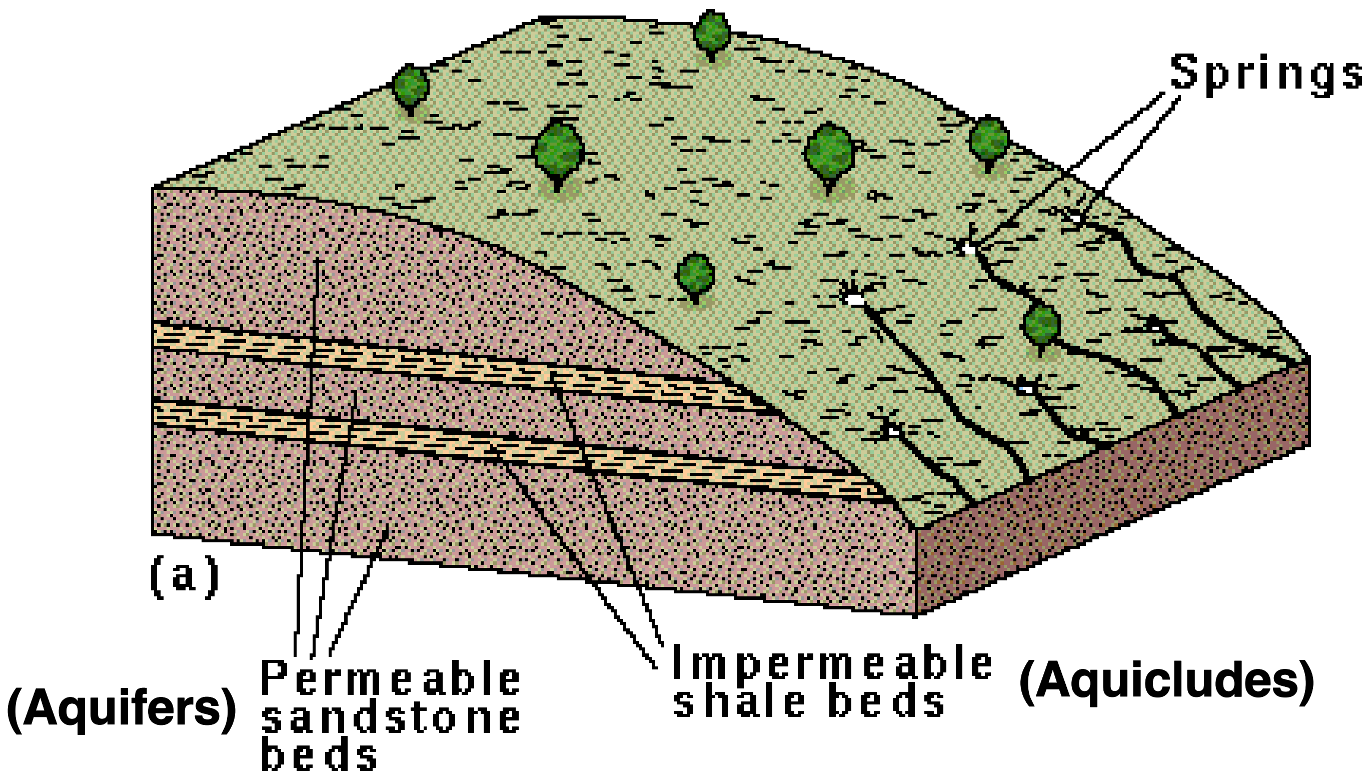
Groundwater velocity increases as the *gradient* increases (steepens).

- Flow velocity of groundwater is variable (very fast to very slow).
- Flows from high to low pressure areas.



Springs

- A place where the water table intersects the ground surface causing water to seep out of the ground.



Hot groundwater-geysers and hot springs

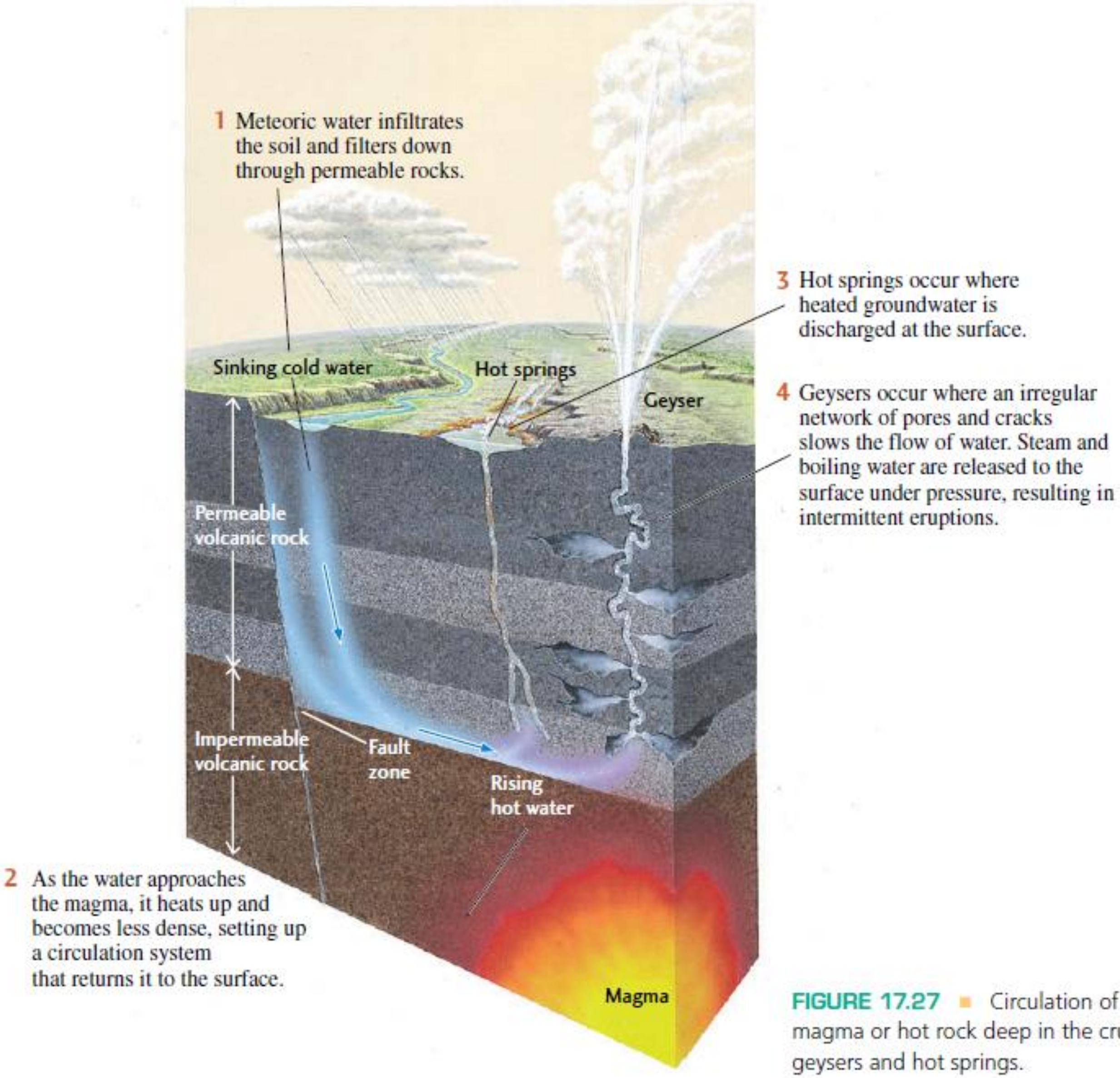


FIGURE 17.27 ■ Circulation of water over magma or hot rock deep in the crust produces geysers and hot springs.

Balancing Recharge and Discharge

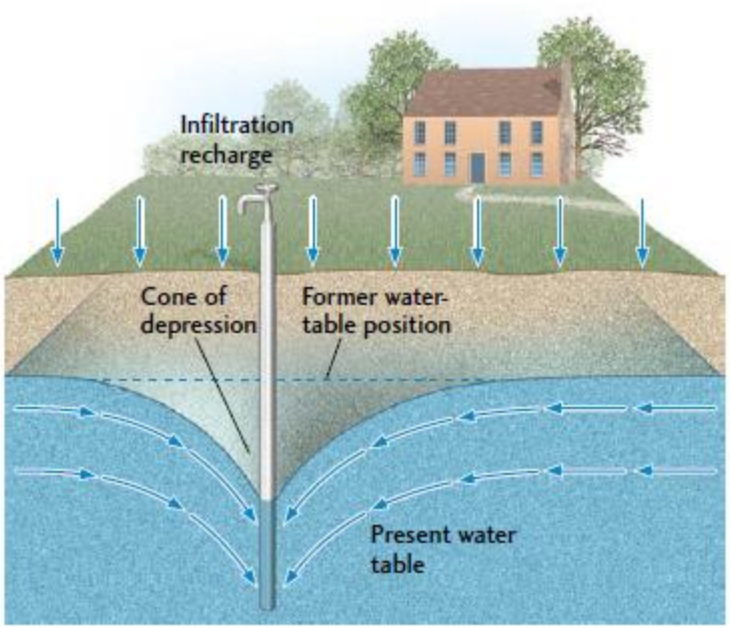
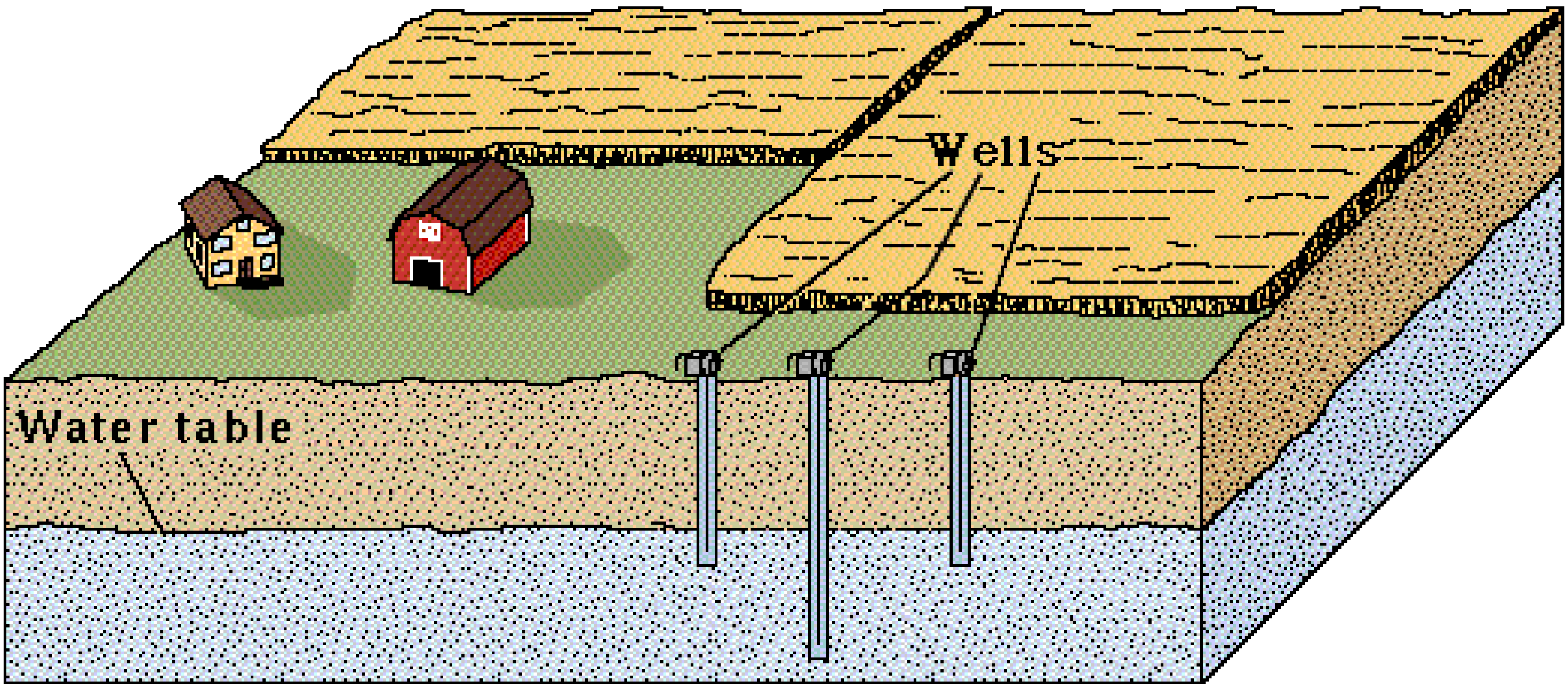


FIGURE 17.15 ■ When discharge from a well exceeds recharge, the water table is lowered in a cone of depression. The water level in the well is lowered to the depressed level of the water table.

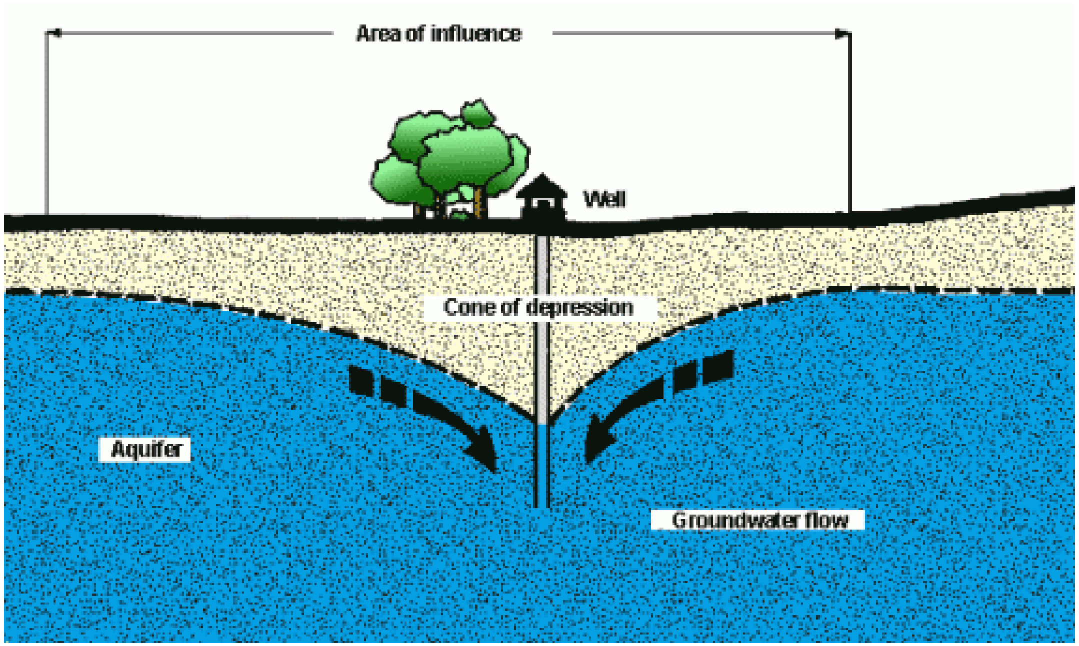
But recharge and discharge are rarely equal because recharge varies with rainfall from season to season. Typically, the water table drops in dry seasons and rises in wet seasons. A longer period of low recharge, such as a prolonged drought, will be followed by a longer-term imbalance and a greater lowering of the water table.

- A water well is made by drilling or digging into the zone of saturation.
- Water pumped from a well causes a cone of depression.



Goundwater overdraft is where the supply cannot replenish as fast as we extract it for human use and results in a decline in water table.

The water table surrounding a well can decline if water is pumped out too fast. The surface of the depleted water table forms a *cone of depression* around the well.



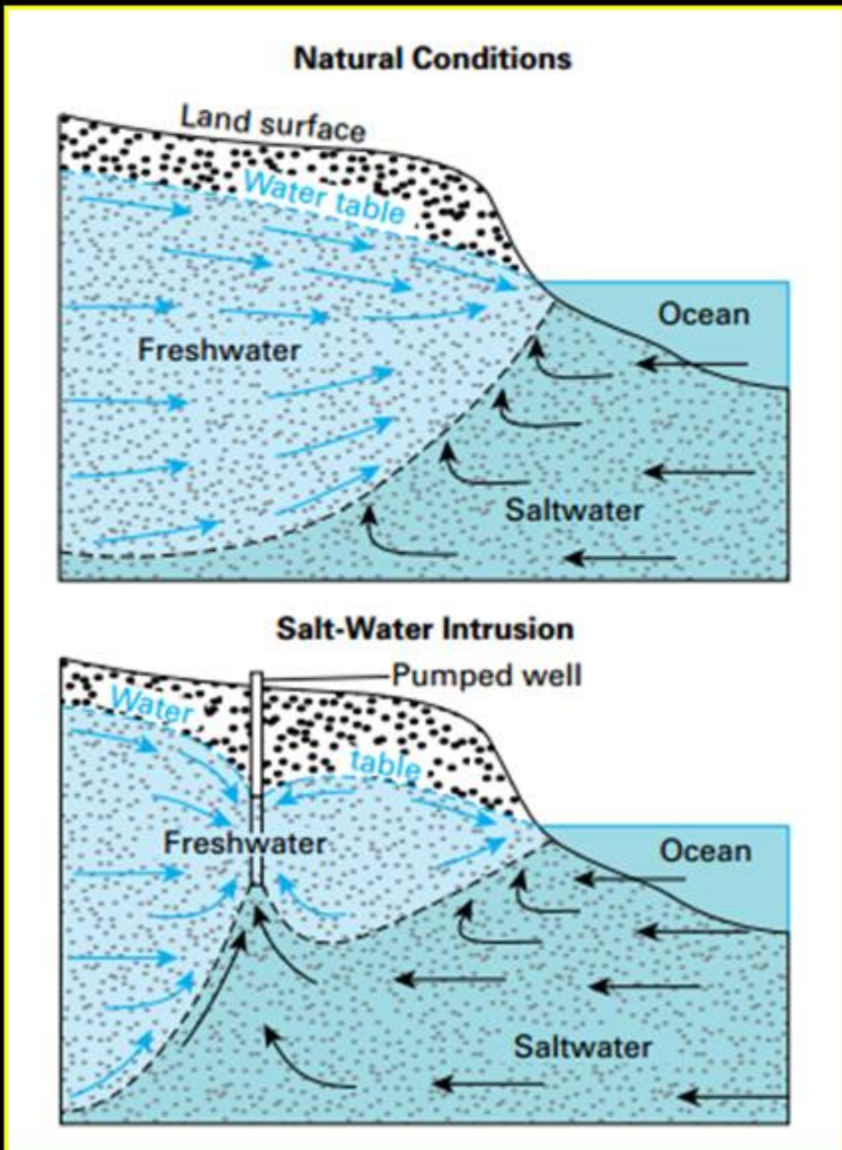
Cornell Cooperative Extension, Cornell University

Rapid population growth in an area usually results in a greater reliance on groundwater as a water source.

Modifications & Effects on Groundwater

- **Currently 1/4th of the world ground water is used by India**

- Over extraction will....
  - 1) lower the water table;
  - 2) decrease hydrostatic pressure;
  - 3) allow saltwater encroachment;
  - 4) cause subsidence;
  - 5) allow contamination of the groundwater.


Natural Conditions
Land surface
Water table
Ocean
Freshwater
Saltwater
Salt-Water Intrusion
Pumped well
Water
table
Freshwater
Ocean
Saltwater

Ground water related subsidence

The 9th century Surang Tila temple in Chhattisgarh has caved in stairs and it is believed to be the result of soil subsidence

Nature can cause contamination of groundwater, but it is commonly due to human activities.

Sources of human and natural contamination can be from *point sources* and *nonpoint sources*.

A *point source* contamination can be specifically identified and located at a specific location such as a leaking gas storage tank or an accident involving chemical tanker trucks and trains.

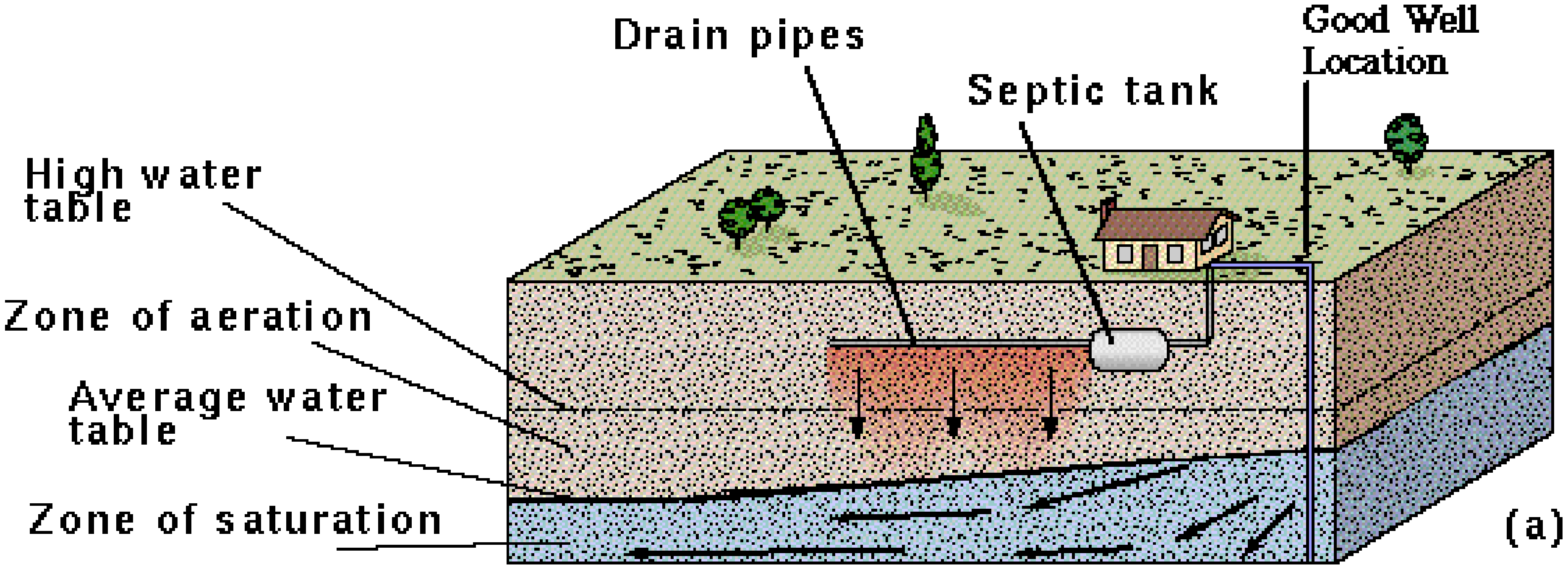
WBMA/abc3340.com /REUTERS

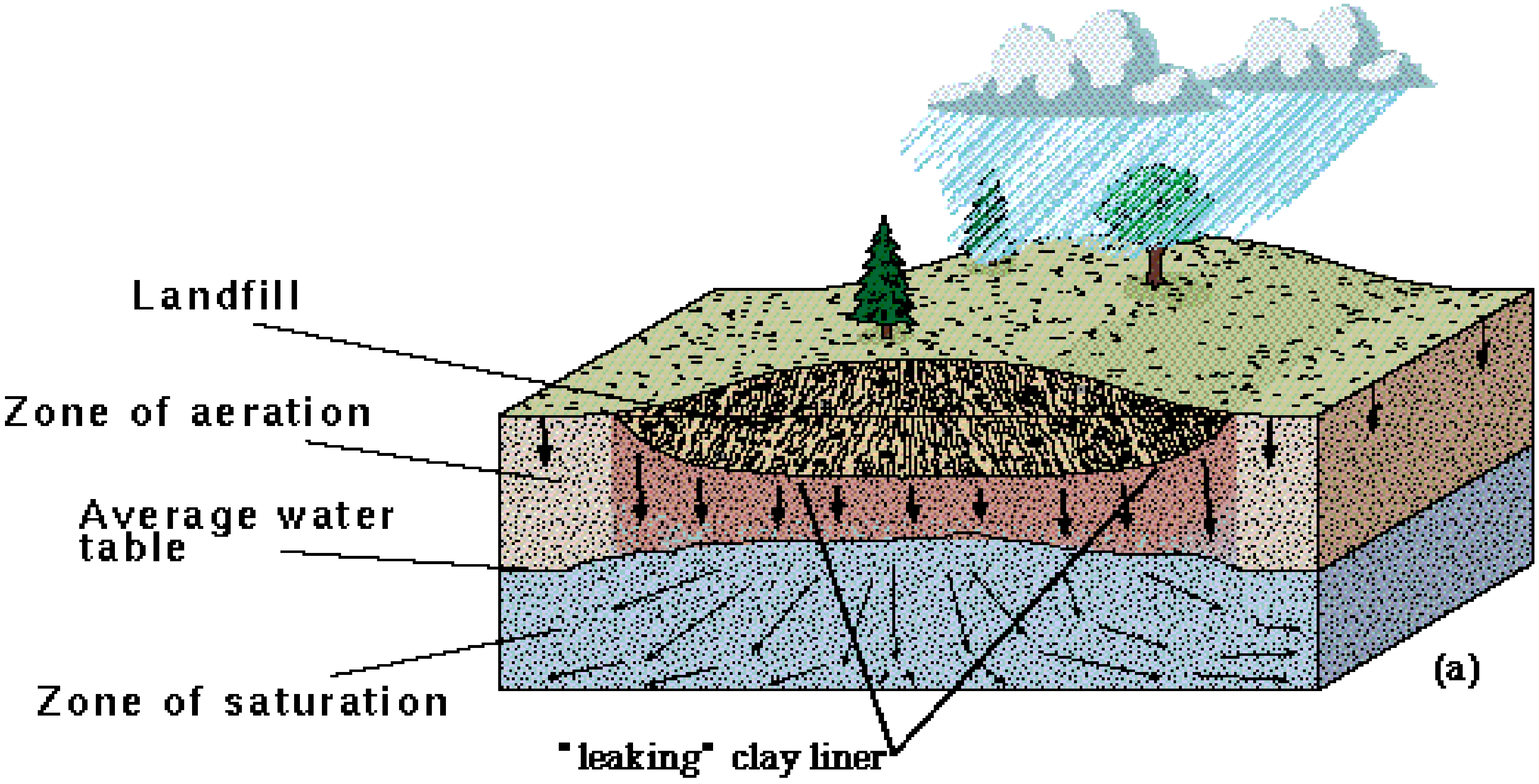
corncrops.com

Nonpoint sources occur over a wide area such as runoff from urban streets and agricultural chemicals (fertilizers and pesticides).

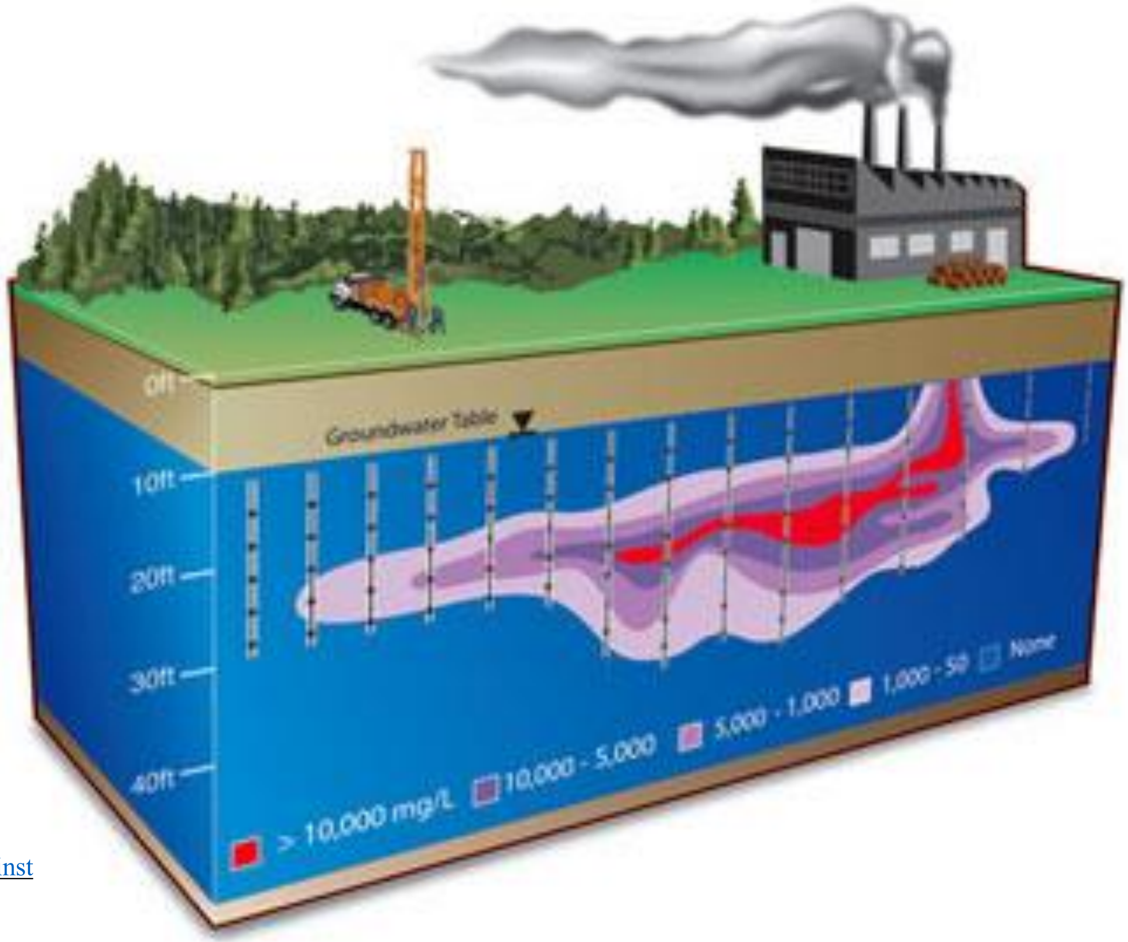
Contaminants can include dissolved metals, organic and inorganic chemical and microbes.

Contamination issues-1: Septic tanks





Contamination issues-2: landfills



When a contaminant enters an aquifer, it can spread as a plume as groundwater flows (due to hydraulic gradient or pumping).

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Groundwater contamination is very difficult to clean up. Methods include

- bioremediation – microbes degrade the contaminant
- groundwater pumping and treatment